

گزارش عیب‌یابی بقای Template های تمرینی ۴، ۵ و ۶ روزه Fitsho

Forced Template Survival + Production Competition Trace

17/30خروجی Competition از خانواده Upper/
Lower**38.3%**

Survival کلی

180

Forced case

36

Template فعال واقعی

نتیجه اصلی: Professional Topology Preference ترتیب تلاش را تغییر می‌دهد، اما survival را تضمین نمی‌کند. در اجرای واقعی این Upper/Lower survival snapshot برابر **68.3%** و سایر topology های دارای امتیاز حرفه‌ای برابر **23.3%** بود.

Generated: 2026-08-30T00:40:38.459860+00:00

HEAD: 9f6f904821891673560bd622ee5ef4b8f56f5cc7

SHA-256-worktree: e12a2f6e1c421da7f1056b5d519e6dd27e0c46c2fb0e5bdde2316797828603c9

SHA-256-raw: ed9b29768fdb5ee30c93119b36d2aa36469e2f513cb175d7bd0cd420775f3eb0

Executive Summary .1

شاخص	مقدار
Template های بررسی شده	36
Forced tests	180
PASS	69
FAIL	111
Success rate	38.3%
Upper/Lower survival	68.3%
Professional topology survival	23.3%
Competition scenarios	30
Upper/Lower competition outputs	17

پنج علت اصلی: (62) RECOVERY_SPACING_INVALID ، (42) SESSION_DURATION_UNDER_TARGET ، (4) TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED ، (2) REQUIRED_MOVEMENT_PATTERN_MISSING ، (1) SESSION_DURATION_TARGET_UNSATISFIED .

پوشش و provenance

کنترل	مقدار
Expected Template × Level × Duration	180
Executed Forced	180
Expected Competition	30
Executed Competition	30
Missing keys	—
Coverage complete	PASS
Active template references — all	53

کنترل	مقدار
Audited references	36
Production exercise catalog	385
Template reference SHA-256	8b19da43901a345c525222a728ba5cf4733268f41bd8a5b7251322a4e82bf33d

Catalog از مسیر واقعی () training_templates.engine_reference.load_template_references() و Exerciseها از مسیر واقعی () WorkoutGenerationService._load_catalog() خوانده شدند. هیچ Template مصنوعی و هیچ seed/migration Audit در اجرا نشد.

Baseline کنترل شده

Gym، همه‌ی Equipment enum ها، بدون recovery context، sleep=good، stress=average، injury/caution/priority/body-analysis، goal=hypertrophy، تمام 30 baseline normalization check دقیقاً به INTERMEDIATE یا ADVANCED مورد انتظار رسیدند.

Catalog واقعی و مشخصات Templateها

p16-4-day-upper-lower-upper-lower-intermediate — بالاتنه / پایین‌تنه / بالاتنه / پایین‌تنه چهار روزه — متوسط

upper_lower=split_type | Upper/Lower=topology | intermediate=days=4 | levels | 4-Day Upper / Lower / Upper / Lower — Intermediate=name
—=specialization_tags | standard=intensity_methods | upper_lower=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	بالاتنه A Upper A	upper	6	3	3	0	chest back shoulders biceps triceps	standard
2	پایین‌تنه A Lower A	lower	5	3	2	0	quadriceps hamstrings glutes calves	standard
3	بالاتنه B Upper B	upper	6	2	4	0	chest back shoulders biceps triceps	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
4	پایین تنه B Lower B	lower	5	2	3	0	quadriceps hamstrings glutes calves	standard

p17-4-day-upper-lower-upper-lower-advanced — بالاتنه / پایین تنه / بالاتنه / پایین تنه چهار روزه — پیشرفته

upper_lower=split_type | Upper/Lower=topology | advanced=days=4 | levels | 4-Day Upper / Lower / Upper / Lower – Advanced=name
—=specialization_tags | standard=intensity_methods | upper_lower=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	بالاتنه A Upper A	upper	6	3	3	0	chest back shoulders biceps triceps	standard
2	پایین تنه A Lower A	lower	5	3	2	0	quadriceps hamstrings glutes calves	standard
3	بالاتنه B Upper B	upper	6	2	4	0	chest back shoulders biceps triceps	standard
4	پایین تنه B Lower B	lower	5	2	3	0	quadriceps hamstrings glutes calves	standard

p19-4-day-3-upper-1-lower-intermediate — چهارروزه؛ سه بالاتنه و یک پایین تنه — متوسط

upper_lower=split_type | Upper/Lower=topology | intermediate=days=4 | levels | 4-Day 3 Upper + 1 Lower – Intermediate=name
—=specialization_tags | standard=intensity_methods | upper_lower=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	بالاتنه A Upper A	upper	6	2	4	0	chest back shoulders biceps triceps	standard
2	پایین‌تنه Lower	lower	5	3	2	0	quadriceps hamstrings glutes calves	standard
3	بالاتنه B Upper B	other	7	1	6	0	chest back shoulders biceps triceps	standard
4	بالاتنه C Upper C	upper	5	2	3	0	chest back shoulders biceps triceps	standard

p20-4-day-3-upper-1-lower-advanced — چهارروزه؛ سه بالاتنه و یک پایین‌تنه — پیشرفته

upper_lower=split_type | Upper/Lower=topology | advanced=days=4 | levels | 4-Day 3 Upper + 1 Lower — Advanced=name
—=specialization_tags | standard=intensity_methods | upper_lower=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	بالاتنه A Upper A	upper	6	2	4	0	chest back shoulders biceps triceps	standard
2	پایین‌تنه Lower	lower	5	3	2	0	quadriceps hamstrings glutes calves	standard
3	بالاتنه B Upper B	other	7	1	6	0	chest back shoulders biceps triceps	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
4	بالاتنه C Upper C	upper	5	2	3	0	chest back shoulders biceps triceps	standard

پ22-4-day-3-lower-1-upper-intermediate — چهارروزه؛ سه پایین‌تنه و یک بالاتنه — متوسط

upper_lower=split_type | Upper/Lower=topology | intermediate=days=4 | levels | 4-Day 3 Lower + 1 Upper – Intermediate=name
upper_lower=focus_tags
lower_priority=specialization_tags | standard=intensity_methods | lower_priority

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	پایین‌تنه A Lower A	lower	4	2	2	0	quadriceps hamstrings glutes calves	standard
2	بالاتنه Upper	upper	8	3	5	0	chest back shoulders biceps triceps	standard
3	پایین‌تنه B Lower B	posterior_chain_core	4	2	2	0	hamstrings glutes	standard
4	پایین‌تنه C Lower C	quadriceps_calves	4	2	2	0	quadriceps glutes	standard

پ23-4-day-3-lower-1-upper-advanced — چهارروزه؛ سه پایین‌تنه و یک بالاتنه — پیشرفته

upper_lower=split_type | Upper/Lower=topology | advanced=days=4 | levels | 4-Day 3 Lower + 1 Upper – Advanced=name
upper_lower=focus_tags
lower_priority=specialization_tags | standard=intensity_methods | lower_priority

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	پایین تنه A Lower A	lower	4	2	2	0	quadriceps hamstrings glutes calves	standard
2	بالا تنه Upper	upper	8	3	5	0	chest back shoulders biceps triceps	standard
3	پایین تنه B Lower B	posterior_chain_core	4	2	2	0	hamstrings glutes	standard
4	پایین تنه C Lower C	quadriceps_calves	4	2	2	0	quadriceps glutes	standard

p24-4-day-push-pull-quads-posterior-intermediate — یوش / پول / چهارسر / خلفی چهارروزه — متوسط

push_pull_legs=split_type | PPL=topology | intermediate=days=4 | levels | 4-Day Push / Pull / Quads / Posterior – Intermediate=name
push_pull_legs=focus_tags
—=specialization_tags | standard=intensity_methods | balanced

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	یوش Push	push	5	2	3	0	chest shoulders triceps	standard
2	پول Pull	pull	4	2	2	0	back biceps traps	standard
3	چهارسر Quads	quadriceps_calves	5	2	3	0	quadriceps calves	standard
4	خلفی Posterior	posterior_chain_core	5	2	3	0	hamstrings glutes	standard

p25-4-day-push-pull-quads-posterior-advanced — پوش / پول / چهارسر / خلفی چهارروزه — پیشرفته

push_pull_legs=split_type | PPL=topology | advanced=days=4 | levels | 4-Day Push / Pull / Quads / Posterior – Advanced=name
push_pull_legs=focus_tags
—=specialization_tags | standard=intensity_methods | balanced

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	پوش Push	push	5	2	3	0	chest shoulders triceps	standard
2	پول Pull	pull	4	2	2	0	back biceps traps	standard
3	چهارسر Quads	quadriceps_calves	5	2	3	0	quadriceps calves	standard
4	خلفی Posterior	posterior_chain_core	5	2	3	0	hamstrings glutes	standard

p50-4-day-iranmuscle-intermediate — ایران ماسل - برنامه ۴ روزه متوسط

body_part_rotation=split_type | Body-Part=topology | intermediate=days=4 | levels | IRANMUSCLE 4-Day Intermediate=name
—=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	جلو پا + ساق Quads + Calves	lower	4	0	4	0	quadriceps calves	standard
2	سینه + پشت بازو Chest + Triceps	upper	5	0	5	0	chest triceps	standard
3	پشت + جلو بازو Back + Biceps	upper	5	0	5	0	back biceps	standard
4	سرشانه + پشت پا Shoulders + Hamstrings	shoulders	5	0	5	0	shoulders hamstrings	standard

p51-4-day-gymextreme-advanced — جیم اکستریم - برنامه ۴ روزه پیشرفته

body_part_rotation=split_type | Body-Part=topology | advanced=days=4 | levels | GymExtreme 4-Day Advanced=name
—=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	پا با تأکید چهارسر Quad-Focused Legs	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard
2	سینه + سرشانه Chest + Shoulders	upper	6	0	6	0	chest shoulders	standard
3	پشت + جلو بازو + پشت بازو Back + Biceps + Triceps	upper	7	0	7	0	back biceps triceps	standard
4	پشت پا + زنجیره خلفی Posterior Chain	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard

p52-4-day-arnoldsho-advanced — آرنولدشو - برنامه ۴ روزه پیشرفته

body_part_rotation=split_type | Arnold-style=topology | advanced=days=4 | levels | Arnoldsho 4-Day Advanced=name
—=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت بازو Chest + Triceps	upper	5	0	5	0	chest triceps	standard
2	پشت + جلو بازو Back + Biceps	upper	6	0	6	0	back biceps	standard
3	سرشانه + کول Shoulders + Traps	shoulders	4	0	4	0	shoulders traps	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
4	پا Legs	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard

p53-4-day-aloplay-intermediate — الویلی - برنامه ۴ روزه متوسط

body_part_rotation=split_type | Body-Part=topology | intermediate=days=4 | levels | Alopplay 4-Day Intermediate=name
—=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت بازو Chest + Triceps	upper	5	0	5	0	chest triceps	standard
2	پا Legs	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
3	پشت + جلو بازو Back + Biceps	upper	5	0	5	0	back biceps	standard
4	سرشانه + شکم Shoulders + Abs	upper	5	0	5	0	shoulders traps abs obliques	standard

p26-5-day-classic-body-part-intermediate — تقسیم کلاسیک عضله‌ای پنج‌روزه — متوسط

body_part_rotation=split_type | Body-Part=topology | intermediate=days=5 | levels | 5-Day Classic Body-Part Split – Intermediate=name
—=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه Chest	chest	4	0	4	0	chest	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
2	پشت Back	back	4	0	4	0	back	standard
3	سرشانه Shoulders	shoulders	4	0	4	0	shoulders traps	standard
4	بازو Arms	arms	5	0	5	0	biceps triceps	standard
5	پا Legs	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard

p27-5-day-classic-body-part-advanced — تقسیم کلاسیک عضله‌ای پنج‌روزه — پیشرفته

body_part_rotation=split_type | Body-Part=topology | advanced=days=5 | levels | 5-Day Classic Body-Part Split – Advanced=name
standard=intensity_methods | body_part_rotation=focus_tags
superset
—=specialization_tags | drop_set

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه Chest	chest	4	0	4	0	chest	standard drop_set
2	پشت Back	back	4	0	4	0	back	standard
3	سرشانه Shoulders	shoulders	4	0	4	0	shoulders traps	standard
4	بازو Arms	arms	5	0	4	0	biceps triceps	standard superset
5	پا Legs	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard

p28-5-day-split-weak-point-intermediate — تقسیم پنج‌روزه با تنوع تمرکز — متوسط

body_part_rotation=split_type | Body-Part=topology | intermediate=days=5 | levels | 5-Day Split + Focus Variation – Intermediate=name
 —=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت بازو Chest + Triceps	upper	5	0	5	0	chest triceps	standard
2	پشت + جلو بازو Back + Biceps	upper	5	0	5	0	back biceps	standard
3	پا Legs	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
4	سرشانه + میان‌تنه Shoulders + Core	upper	5	0	5	0	shoulders traps	standard
5	نقطه ضعف / تمام‌بدن سبک Weak Point / Light Full Body	full_body	5	0	5	0	quadriceps hamstrings glutes calves chest back shoulders biceps triceps	standard

p29-5-day-split-weak-point-advanced — تقسیم پنج‌روزه با تنوع تمرکز — پیشرفته

body_part_rotation=split_type | Body-Part=topology | advanced=days=5 | levels | 5-Day Split + Focus Variation – Advanced=name
 standard=intensity_methods | body_part_rotation=focus_tags
 —=specialization_tags | drop_set

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت بازو Chest + Triceps	upper	5	0	5	0	chest triceps	standard
2	پشت + جلو بازو Back + Biceps	upper	5	0	5	0	back biceps	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
3	پا Legs	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
4	سرشانه + میان تنه Shoulders + Core	upper	5	0	5	0	shoulders traps	standard
5	نقطه ضعف / تمام بدن سبک Weak Point / Light Full Body	full_body	5	0	5	0	quadriceps hamstrings glutes calves chest back shoulders biceps triceps	standard drop_set

p30-5-day-upper-priority-iranian-intermediate — تقسیم ایرانی پنج روزه بالاتنه / پایین تنه — متوسط

upper_lower=split_type | Upper/Lower=topology | intermediate=days=5 | levels | 5-Day Upper / Lower Iranian Split – Intermediate=name
—=specialization_tags | standard=intensity_methods | upper_lower=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت بازو Chest + Triceps	upper	5	0	5	0	chest back shoulders biceps triceps	standard
2	سرشانه + جلو بازو Shoulders + Biceps	upper	5	0	5	0	chest back shoulders biceps triceps	standard
3	پا + میان تنه Legs + Core	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
4	بالاسینه + جلو بازو Upper Chest + Biceps	upper	5	0	5	0	chest back shoulders biceps triceps	standard
5	پشت + میان تنه Back + Core	upper	5	0	5	0	chest back shoulders biceps triceps	standard

— تقسیم ایرانی پنج روزه بالاتنه / پایین تنه — پیشرفته

upper_lower=split_type | Upper/Lower=topology | advanced=days=5 | levels | 5-Day Upper / Lower Iranian Split – Advanced=name
standard=intensity_methods | upper_lower=focus_tags
—=specialization_tags | drop_set

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت بازو Chest + Triceps	upper	5	0	5	0	chest back shoulders biceps triceps	standard
2	سرشانه + جلو بازو Shoulders + Biceps	upper	5	0	5	0	chest back shoulders biceps triceps	standard
3	پا + میان تنه Legs + Core	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
4	بالاسینه + جلو بازو Upper Chest + Biceps	upper	5	0	5	0	chest back shoulders biceps triceps	standard drop_set

Methods	Target muscles	Optional	Accessory	Core slots	Initial exercises	structure_focus	عنوان	روز
standard	chest back shoulders biceps triceps	0	5	0	5	upper	پشت + میان تنه Back + Core	5

p32-5-day-upper-lower-specialty-intermediate — تنوع پنج روزه بالاتنه / پایین تنه — متوسط

upper_lower=split_type | Upper/Lower=topology | intermediate=days=5 | levels | 5-Day Upper / Lower Variation – Intermediate=name
—=specialization_tags | standard=intensity_methods | upper_lower=focus_tags

Methods	Target muscles	Optional	Accessory	Core slots	Initial exercises	structure_focus	عنوان	روز
standard	chest back shoulders biceps triceps	0	6	0	6	upper	بالاتنه A Upper A	1
standard	quadriceps hamstrings glutes calves	0	5	0	5	lower	پایین تنه A Lower A	2
standard	chest back shoulders biceps triceps	0	6	0	6	upper	بالاتنه B Upper B	3
standard	quadriceps hamstrings glutes calves	0	5	0	5	lower	پایین تنه B Lower B	4
standard	shoulders biceps triceps	0	6	0	6	upper	تخصص بازو + دلت Arms + Delts Specialty	5

p33-5-day-upper-lower-specialty-advanced — تنوع پنج‌روزه بالاتنه / پایین‌تنه — پیشرفته

upper_lower=split_type | Upper/Lower=topology | advanced=days=5 | levels | 5-Day Upper / Lower Variation – Advanced=name
 standard=intensity_methods | upper_lower=focus_tags
 superset
 —=specialization_tags | drop_set

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	بالته A Upper A	upper	4	0	4	0	chest back shoulders biceps triceps	standard
2	پایین‌ته A Lower A	lower	4	0	4	0	quadriceps hamstrings glutes calves	standard
3	بالته B Upper B	upper	5	0	5	0	chest back shoulders biceps triceps	standard
4	پایین‌ته B Lower B	lower	4	0	4	0	quadriceps hamstrings glutes calves	standard
5	تخصص بازو + دلت Arms + Delts Specialty	upper	6	0	5	0	shoulders biceps triceps	standard drop_set superset

p34-5-day-fst7-arms-priority-intermediate — پنج‌روزه FST-7 با اولویت بازو — متوسط

body_part_rotation=split_type | Specialization=topology | intermediate=days=5 | levels | 5-Day FST-7 / Arms Priority – Intermediate=name
 body_part_rotation=focus_tags
 arms_priority
 arms_priority=specialization_tags | standard=intensity_methods | specialization
 specialization

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + جلو بازو Chest + Biceps	upper	5	0	5	0	chest back shoulders biceps triceps	standard
2	پشت + پشت بازو Back + Triceps	upper	5	0	5	0	chest back shoulders biceps triceps	standard
3	پا Legs	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard
4	سرشانه + ساق Shoulders + Calves	shoulders	5	0	5	0	shoulders traps	standard
5	بازو Arms	arms	5	0	5	0	biceps triceps	standard

پنج‌روزه FST-7 با اولویت بازو — پیشرفته p35-5-day-fst7-arms-priority-advanced

body_part_rotation=split_type | Specialization=topology | advanced=days=5 | levels | 5-Day FST-7 / Arms Priority – Advanced=name
body_part_rotation=focus_tags
arms_priority
arms_priority=specialization_tags | standard=intensity_methods | specialization
specialization

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + جلو بازو Chest + Biceps	upper	4	0	4	0	chest back shoulders biceps triceps	standard
2	پشت + پشت بازو Back + Triceps	upper	4	0	4	0	chest back shoulders biceps triceps	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
3	پا Legs	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
4	سرشانه + ساق Shoulders + Calves	shoulders	4	0	4	0	shoulders traps	standard
5	بازو Arms	arms	4	0	4	0	biceps triceps	standard

p36-5-day-professional-compound-intermediate — تقسیم حرفه‌ای پنج‌روزه با روز ترکیبی — متوسط

body_part_rotation=split_type | Body-Part=topology | intermediate=days=5 | levels | 5-Day Professional Split + Compound Day – Intermediate=name
—=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت بازو Chest + Triceps	upper	4	0	4	0	chest back shoulders biceps triceps	standard
2	پا + میان‌تنه Legs + Core	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
3	پشت + جلو بازو Back + Biceps	upper	5	0	5	0	chest back shoulders biceps triceps	standard
4	سرشانه Shoulders	shoulders	4	0	4	0	shoulders traps	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
5	روز حرکات ترکیبی Compound Day	full_body	5	0	5	0	quadriceps hamstrings glutes calves chest back shoulders biceps triceps	standard

p37-5-day-professional-compound-advanced — تقسیم حرفه‌ای پنج‌روزه با روز ترکیبی — پیشرفته

body_part_rotation=split_type | Body-Part=topology | advanced=days=5 | levels | 5-Day Professional Split + Compound Day – Advanced=name
—=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت بازو Chest + Triceps	upper	4	0	4	0	chest back shoulders biceps triceps	standard
2	پا + میان‌تنه Legs + Core	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
3	پشت + جلو بازو Back + Biceps	upper	5	0	5	0	chest back shoulders biceps triceps	standard
4	سرشانه Shoulders	shoulders	4	0	4	0	shoulders traps	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
5	روز حرکات ترکیبی Compound Day	full_body	5	0	5	0	quadriceps hamstrings glutes calves chest back shoulders biceps triceps	standard

p38-6-day-ppl-ab-intermediate — پوش / پول / پا A/B — متوسط

push_pull_legs=split_type | PPLx2=topology | intermediate=days=6 | levels | PPL A/B – Intermediate=name
—=specialization_tags | standard=intensity_methods | push_pull_legs=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	پوش A Push A	push	5	0	5	0	chest shoulders triceps	standard
2	پول A Pull A	pull	5	0	5	0	back biceps traps	standard
3	پا A Legs A	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
4	پوش B Push B	push	5	0	5	0	chest shoulders triceps	standard
5	پول B Pull B	pull	5	0	5	0	back biceps shoulders	standard
6	پا B Legs B	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard

p39-6-day-pp1-ab-advanced — پوش / پول / یا A/B — پیشرفته

push_pull_legs=split_type | PPLx2=topology | advanced=days=6 | levels | PPL A/B – Advanced=name
standard=intensity_methods | push_pull_legs=focus_tags
—=specialization_tags | drop_set

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	پوش A Push A	push	5	0	5	0	chest shoulders triceps	standard
2	پول A Pull A	pull	5	0	5	0	back biceps traps	standard
3	پا A Legs A	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard drop_set
4	پوش B Push B	push	5	0	5	0	chest shoulders triceps	standard
5	پول B Pull B	pull	5	0	5	0	back biceps shoulders	standard
6	پا B Legs B	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard

p40-6-day-upper-lower-x3-intermediate — بالاتنه / پایین‌تنه سه‌بار در هفته — متوسط

upper_lower=split_type | Upper/Lower=topology | intermediate=days=6 | levels | 6-Day Upper / Lower x3 – Intermediate=name
—=specialization_tags | standard=intensity_methods | upper_lower=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	بالاتنه A Upper A	upper	6	0	6	0	chest back shoulders biceps triceps	standard
2	پایین‌تنه A Lower A	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
3	بالاتنه B Upper B	upper	6	0	6	0	chest back shoulders biceps triceps	standard
4	پایین‌تنه B Lower B	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
5	بالاتنه C Upper C	upper	6	0	6	0	chest back shoulders biceps triceps	standard
6	پایین‌تنه C Lower C	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard

p41-6-day-upper-lower-x3-advanced — بالاتنه / پایین‌تنه سه‌بار در هفته — پیشرفته

upper_lower=split_type | Upper/Lower=topology | advanced=days=6 | levels | 6-Day Upper / Lower x3 – Advanced=name
standard=intensity_methods | upper_lower=focus_tags
superset
—=specialization_tags | drop_set

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	بالاتنه A Upper A	upper	4	0	4	0	chest back shoulders biceps triceps	standard
2	پایین‌تنه A Lower A	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
3	بالاتنه B Upper B	upper	6	0	6	0	chest back shoulders biceps triceps	standard drop_set
4	پایین‌تنه B Lower B	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
5	بالاتنه C Upper C	upper	6	0	5	0	chest back shoulders biceps triceps	standard superset
6	پایین‌تنه C Lower C	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard

p42-6-day-fitclub-hybrid-intermediate — ترکیبی شش‌روزه فیت‌کلاب — متوسط

push_pull_legs=split_type | PPL=topology | intermediate=days=6 | levels | 6-Day FitClub Hybrid – Intermediate=name
—=specialization_tags | standard=intensity_methods | push_pull_legs=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت بازو Chest + Triceps	upper	5	0	5	0	chest triceps	standard
2	پشت + جلو بازو Back + Biceps	upper	5	0	5	0	back biceps	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
3	پا Legs	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
4	سرشانه + میان تنه Shoulders + Core	shoulders	5	0	5	0	shoulders traps	standard
5	سینه + پشت Chest + Back	upper	6	0	6	0	chest back	standard
6	خلفی + میان تنه Posterior + Core	lower	6	0	6	0	hamstrings glutes calves	standard

p43-6-day-fitclub-hybrid-advanced — ترکیبی شش روزه فیت کلاب — پیشرفته

push_pull_legs=split_type | PPL=topology | advanced=days=6 | levels | 6-Day FitClub Hybrid – Advanced=name
 standard=intensity_methods | push_pull_legs=focus_tags
 —=specialization_tags | superset

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت بازو Chest + Triceps	upper	5	0	5	0	chest triceps	standard
2	پشت + جلو بازو Back + Biceps	upper	5	0	5	0	back biceps	standard
3	پا Legs	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
4	سرشانه + میان تنه Shoulders + Core	shoulders	5	0	5	0	shoulders traps	standard
5	سینه + پشت Chest + Back	upper	6	0	5	0	chest back	standard superset
6	خلفی + میان تنه Posterior + Core	lower	6	0	6	0	hamstrings glutes calves	standard

— تقسیم آرنولد شش‌روزه — متوسط p44-6-day-arnold-split-intermediate

body_part_rotation=split_type | Arnold-style=topology | intermediate=days=6 | levels | 6-Day Arnold Split – Intermediate=name
—=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت A Chest + Back A	upper	6	0	6	0	chest back	standard
2	سرشانه + بازو A Shoulders + Arms A	upper	6	0	6	0	shoulders biceps triceps	standard
3	پا A Legs A	lower	4	0	4	0	quadriceps hamstrings glutes calves	standard
4	سینه + پشت B Chest + Back B	upper	6	0	6	0	chest back	standard
5	سرشانه + بازو B Shoulders + Arms B	upper	6	0	6	0	shoulders biceps triceps	standard
6	پا B Legs B	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard

— تقسیم آرنولد شش‌روزه — پیشرفته p45-6-day-arnold-split-advanced

body_part_rotation=split_type | Arnold-style=topology | advanced=days=6 | levels | 6-Day Arnold Split – Advanced=name
standard=intensity_methods | body_part_rotation=focus_tags
—=specialization_tags | superset

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه + پشت A Chest + Back A	upper	6	0	4	0	chest back	standard superset
2	سرشانه + بازو A Shoulders + Arms A	upper	5	0	4	0	shoulders biceps triceps	standard superset

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
3	پا A Legs A	lower	4	0	4	0	quadriceps hamstrings glutes calves	standard
4	سینه + پشت B Chest + Back B	upper	6	0	5	0	chest back	standard superset
5	سرشانه + بازو B Shoulders + Arms B	upper	5	0	4	0	shoulders biceps triceps	standard superset
6	پا B Legs B	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard

p46-6-day-classic-body-part-intermediate — تقسیم کلاسیک شش‌روزه عضله‌ای — متوسط

body_part_rotation=split_type | Body-Part=topology | intermediate=days=6 | levels | 6-Day Classic Six Body-Part – Intermediate=name
—=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه Chest	chest	4	0	4	0	chest	standard
2	جلو بازو Biceps	arms	4	0	4	0	biceps	standard
3	پا Legs	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
4	پشت بازو Triceps	arms	3	0	3	0	triceps	standard
5	پشت Back	back	5	0	5	0	back	standard
6	سرشانه Shoulders	shoulders	4	0	4	0	shoulders traps	standard

p47-6-day-classic-body-part-advanced — تقسیم کلاسیک شش‌روزه عضله‌ای — پیشرفته

body_part_rotation=split_type | Body-Part=topology | advanced=days=6 | levels | 6-Day Classic Six Body-Part – Advanced=name
standard=intensity_methods | body_part_rotation=focus_tags
—=specialization_tags | drop_set

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	سینه Chest	chest	4	0	4	0	chest	standard drop_set
2	جلو بازو Biceps	arms	4	0	4	0	biceps	standard
3	پا Legs	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
4	پشت بازو Triceps	arms	3	0	3	0	triceps	standard
5	پشت Back	back	4	0	4	0	back	standard
6	سرشانه Shoulders	shoulders	4	0	4	0	shoulders traps	standard

p48-6-day-ronnie-double-exposure-intermediate — مواجهه دوگانه رانی شش‌روزه — متوسط

body_part_rotation=split_type | Body-Part=topology | intermediate=days=6 | levels | 6-Day Ronnie Double Exposure – Intermediate=name
—=specialization_tags | standard=intensity_methods | body_part_rotation=focus_tags

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	پشت + جلو بازو + سرشانه A Back + Biceps + Shoulders A	upper	5	0	5	0	back biceps shoulders	standard
2	پا A Legs A	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
3	سینه + پشت بازو A Chest + Triceps A	upper	5	0	5	0	chest triceps	standard
4	پشت + جلو بازو + سرشانه B Back + Biceps + Shoulders B	upper	5	0	5	0	back biceps shoulders	standard
5	پا B Legs B	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard
6	سینه + پشت بازو B Chest + Triceps B	upper	5	0	5	0	chest triceps	standard

p49-6-day-ronnie-double-exposure-advanced — مواجهه دوگانه رانی شش‌روزه — پیشرفته

body_part_rotation=split_type | Body-Part=topology | advanced=days=6 | levels | 6-Day Ronnie Double Exposure – Advanced=name
standard=intensity_methods | body_part_rotation=focus_tags
—=specialization_tags | drop_set

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
1	پشت + جلو بازو + سرشانه A Back + Biceps + Shoulders A	upper	5	0	5	0	back biceps shoulders	standard
2	پا A Legs A	lower	5	0	5	0	quadriceps hamstrings glutes calves	standard
3	سینه + پشت بازو A Chest + Triceps A	upper	5	0	5	0	chest triceps	standard
4	پشت + جلو بازو + سرشانه B Back + Biceps + Shoulders B	upper	6	0	6	0	back biceps shoulders	standard drop_set
5	پا B Legs B	lower	6	0	6	0	quadriceps hamstrings glutes calves	standard

روز	عنوان	structure_focus	Initial exercises	Core slots	Accessory	Optional	Target muscles	Methods
6	سینه + پشت بازو B Chest + Triceps B	upper	5	0	5	0	chest triceps	standard

Test Matrix .2

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED	p50-4-day-iranmuscle-intermediate	SUCCESS	FAIL	30	4	intermediate	Upper/Lower	p16-4-day-upper-lower-upper-lower-intermediate
–	p50-4-day-iranmuscle-intermediate	SUCCESS	PASS	45	4	intermediate	Upper/Lower	p16-4-day-upper-lower-upper-lower-intermediate
–	p53-4-day-aloplay-intermediate	SUCCESS	PASS	60	4	intermediate	Upper/Lower	p16-4-day-upper-lower-upper-lower-intermediate
–	p53-4-day-aloplay-intermediate	SUCCESS	PASS	75	4	intermediate	Upper/Lower	p16-4-day-upper-lower-upper-lower-intermediate
–	p50-4-day-iranmuscle-intermediate	SUCCESS	PASS	90	4	intermediate	Upper/Lower	p16-4-day-upper-lower-upper-lower-intermediate
TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED	p52-4-day-arnoldsho-advanced	SUCCESS	FAIL	30	4	advanced	Upper/Lower	p17-4-day-upper-lower-upper-lower-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	45	4	advanced	Upper/Lower	p17-4-day-upper-lower-upper-lower-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	60	4	advanced	Upper/Lower	p17-4-day-upper-lower-upper-lower-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	75	4	advanced	Upper/Lower	p17-4-day-upper-lower-upper-lower-advanced
–	p51-4-day-gymextreme-advanced	SUCCESS	PASS	90	4	advanced	Upper/Lower	p17-4-day-upper-lower-upper-lower-advanced
–	p50-4-day-iranmuscle-intermediate	SUCCESS	PASS	30	4	intermediate	Upper/Lower	p19-4-day-3-upper-1-lower-intermediate
–	p50-4-day-iranmuscle-intermediate	SUCCESS	PASS	45	4	intermediate	Upper/Lower	p19-4-day-3-upper-1-lower-intermediate

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
–	p53-4-day-alo play-intermediate	SUCCESS	PASS	60	4	intermediate	Upper/Lower	p19-4-day-3-upper-1-lower-intermediate
–	p53-4-day-alo play-intermediate	SUCCESS	PASS	75	4	intermediate	Upper/Lower	p19-4-day-3-upper-1-lower-intermediate
–	p50-4-day-iran muscle-intermediate	SUCCESS	PASS	90	4	intermediate	Upper/Lower	p19-4-day-3-upper-1-lower-intermediate
–	p52-4-day-arnold sho-advanced	SUCCESS	PASS	30	4	advanced	Upper/Lower	p20-4-day-3-upper-1-lower-advanced
–	p52-4-day-arnold sho-advanced	SUCCESS	PASS	45	4	advanced	Upper/Lower	p20-4-day-3-upper-1-lower-advanced
–	p52-4-day-arnold sho-advanced	SUCCESS	PASS	60	4	advanced	Upper/Lower	p20-4-day-3-upper-1-lower-advanced
–	p52-4-day-arnold sho-advanced	SUCCESS	PASS	75	4	advanced	Upper/Lower	p20-4-day-3-upper-1-lower-advanced
–	p51-4-day-gym extreme-advanced	SUCCESS	PASS	90	4	advanced	Upper/Lower	p20-4-day-3-upper-1-lower-advanced
TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED	p50-4-day-iran muscle-intermediate	SUCCESS	FAIL	30	4	intermediate	Upper/Lower	p22-4-day-3-lower-1-upper-intermediate
–	p50-4-day-iran muscle-intermediate	SUCCESS	PASS	45	4	intermediate	Upper/Lower	p22-4-day-3-lower-1-upper-intermediate
–	p53-4-day-alo play-intermediate	SUCCESS	PASS	60	4	intermediate	Upper/Lower	p22-4-day-3-lower-1-upper-intermediate
–	p53-4-day-alo play-intermediate	SUCCESS	PASS	75	4	intermediate	Upper/Lower	p22-4-day-3-lower-1-upper-intermediate
SESSION_DURATION_UNDER_TARGET	p50-4-day-iran muscle-intermediate	SUCCESS	FAIL	90	4	intermediate	Upper/Lower	p22-4-day-3-lower-1-upper-intermediate
TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED	p52-4-day-arnold sho-advanced	SUCCESS	FAIL	30	4	advanced	Upper/Lower	p23-4-day-3-lower-1-upper-advanced

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	45	4	advanced	Upper/Lower	p23-4-day-3-lower-1-upper-advanced
SESSION_DURATION_TARGET_UNSATISFIED	p52-4-day-arnoldsho-advanced	SUCCESS	FAIL	60	4	advanced	Upper/Lower	p23-4-day-3-lower-1-upper-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	75	4	advanced	Upper/Lower	p23-4-day-3-lower-1-upper-advanced
SESSION_DURATION_UNDER_TARGET	p51-4-day-gymextreme-advanced	SUCCESS	FAIL	90	4	advanced	Upper/Lower	p23-4-day-3-lower-1-upper-advanced
–	p50-4-day-iranmuscle-intermediate	SUCCESS	PASS	30	4	intermediate	PPL	p24-4-day-push-pull-quads-posterior-intermediate
–	p50-4-day-iranmuscle-intermediate	SUCCESS	PASS	45	4	intermediate	PPL	p24-4-day-push-pull-quads-posterior-intermediate
–	p53-4-day-aloplay-intermediate	SUCCESS	PASS	60	4	intermediate	PPL	p24-4-day-push-pull-quads-posterior-intermediate
–	p53-4-day-aloplay-intermediate	SUCCESS	PASS	75	4	intermediate	PPL	p24-4-day-push-pull-quads-posterior-intermediate
–	p50-4-day-iranmuscle-intermediate	SUCCESS	PASS	90	4	intermediate	PPL	p24-4-day-push-pull-quads-posterior-intermediate
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	30	4	advanced	PPL	p25-4-day-push-pull-quads-posterior-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	45	4	advanced	PPL	p25-4-day-push-pull-quads-posterior-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	60	4	advanced	PPL	p25-4-day-push-pull-quads-posterior-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	75	4	advanced	PPL	p25-4-day-push-pull-quads-posterior-advanced

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
–	p51-4-day-gymextreme-advanced	SUCCESS	PASS	90	4	advanced	PPL	p25-4-day-push-pull-quads-posterior-advanced
–	p50-4-day-iranmuscle-intermediate	SUCCESS	PASS	30	4	intermediate	Body-Part	p50-4-day-iranmuscle-intermediate
–	p50-4-day-iranmuscle-intermediate	SUCCESS	PASS	45	4	intermediate	Body-Part	p50-4-day-iranmuscle-intermediate
–	p53-4-day-aloplay-intermediate	SUCCESS	PASS	60	4	intermediate	Body-Part	p50-4-day-iranmuscle-intermediate
–	p53-4-day-aloplay-intermediate	SUCCESS	PASS	75	4	intermediate	Body-Part	p50-4-day-iranmuscle-intermediate
–	p50-4-day-iranmuscle-intermediate	SUCCESS	PASS	90	4	intermediate	Body-Part	p50-4-day-iranmuscle-intermediate
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	30	4	advanced	Body-Part	p51-4-day-gymextreme-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	45	4	advanced	Body-Part	p51-4-day-gymextreme-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	60	4	advanced	Body-Part	p51-4-day-gymextreme-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	75	4	advanced	Body-Part	p51-4-day-gymextreme-advanced
–	p51-4-day-gymextreme-advanced	SUCCESS	PASS	90	4	advanced	Body-Part	p51-4-day-gymextreme-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	30	4	advanced	Arnold-style	p52-4-day-arnoldsho-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	45	4	advanced	Arnold-style	p52-4-day-arnoldsho-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	60	4	advanced	Arnold-style	p52-4-day-arnoldsho-advanced
–	p52-4-day-arnoldsho-advanced	SUCCESS	PASS	75	4	advanced	Arnold-style	p52-4-day-arnoldsho-advanced

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
SESSION_DURATION_UNDER_TARGET	p51-4-day-gymextreme-advanced	SUCCESS	FAIL	90	4	advanced	Arnold-style	p52-4-day-arnoldsho-advanced
REQUIRED_MOVEMENT_PATTERN_MISSING	p50-4-day-iranmuscle-intermediate	SUCCESS	FAIL	30	4	intermediate	Body-Part	p53-4-day-aloplay-intermediate
REQUIRED_MOVEMENT_PATTERN_MISSING	p50-4-day-iranmuscle-intermediate	SUCCESS	FAIL	45	4	intermediate	Body-Part	p53-4-day-aloplay-intermediate
–	p53-4-day-aloplay-intermediate	SUCCESS	PASS	60	4	intermediate	Body-Part	p53-4-day-aloplay-intermediate
–	p53-4-day-aloplay-intermediate	SUCCESS	PASS	75	4	intermediate	Body-Part	p53-4-day-aloplay-intermediate
SESSION_DURATION_UNDER_TARGET	p50-4-day-iranmuscle-intermediate	SUCCESS	FAIL	90	4	intermediate	Body-Part	p53-4-day-aloplay-intermediate
–	p26-5-day-classic-body-part-intermediate	SUCCESS	PASS	30	5	intermediate	Body-Part	p26-5-day-classic-body-part-intermediate
RECOVERY_SPACING_INVALID	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	45	5	intermediate	Body-Part	p26-5-day-classic-body-part-intermediate
SESSION_DURATION_UNDER_TARGET	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	60	5	intermediate	Body-Part	p26-5-day-classic-body-part-intermediate
SESSION_DURATION_UNDER_TARGET	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	75	5	intermediate	Body-Part	p26-5-day-classic-body-part-intermediate
SESSION_DURATION_UNDER_TARGET	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	90	5	intermediate	Body-Part	p26-5-day-classic-body-part-intermediate
RECOVERY_SPACING_INVALID	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	30	5	advanced	Body-Part	p27-5-day-classic-body-part-advanced
RECOVERY_SPACING_INVALID	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	45	5	advanced	Body-Part	p27-5-day-classic-body-part-advanced
RECOVERY_SPACING_INVALID	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	60	5	advanced	Body-Part	p27-5-day-classic-body-part-advanced

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
SESSION_DURATION_UPPER_TARGET	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	75	5	advanced	Body-Part	p27-5-day-classic-body-part-advanced
SESSION_DURATION_UPPER_TARGET	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	90	5	advanced	Body-Part	p27-5-day-classic-body-part-advanced
RECOVERY_SPACING_INVALID	p26-5-day-classic-body-part-intermediate	SUCCESS	FAIL	30	5	intermediate	Body-Part	p28-5-day-split-weak-point-intermediate
RECOVERY_SPACING_INVALID	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	45	5	intermediate	Body-Part	p28-5-day-split-weak-point-intermediate
RECOVERY_SPACING_INVALID	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	60	5	intermediate	Body-Part	p28-5-day-split-weak-point-intermediate
SESSION_DURATION_UPPER_TARGET	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	75	5	intermediate	Body-Part	p28-5-day-split-weak-point-intermediate
SESSION_DURATION_UPPER_TARGET	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	90	5	intermediate	Body-Part	p28-5-day-split-weak-point-intermediate
RECOVERY_SPACING_INVALID	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	30	5	advanced	Body-Part	p29-5-day-split-weak-point-advanced
RECOVERY_SPACING_INVALID	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	45	5	advanced	Body-Part	p29-5-day-split-weak-point-advanced
RECOVERY_SPACING_INVALID	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	60	5	advanced	Body-Part	p29-5-day-split-weak-point-advanced
RECOVERY_SPACING_INVALID	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	75	5	advanced	Body-Part	p29-5-day-split-weak-point-advanced
SESSION_DURATION_UPPER_TARGET	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	90	5	advanced	Body-Part	p29-5-day-split-weak-point-advanced
RECOVERY_SPACING_INVALID	p26-5-day-classic-body-part-intermediate	SUCCESS	FAIL	30	5	intermediate	Upper/Lower	p30-5-day-upper-priority-iranian-intermediate
RECOVERY_SPACING_INVALID	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	45	5	intermediate	Upper/Lower	p30-5-day-upper-priority-iranian-intermediate

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
RECOVERY_SPACING_I NVALID	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	FAIL	60	5	intermediate	Upper/Lower	p30-5-day-upper- priority-iranian- intermediate
RECOVERY_SPACING_I NVALID	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	FAIL	75	5	intermediate	Upper/Lower	p30-5-day-upper- priority-iranian- intermediate
SESSION_DURATION_U NDER_TARGET	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	FAIL	90	5	intermediate	Upper/Lower	p30-5-day-upper- priority-iranian- intermediate
RECOVERY_SPACING_I NVALID	p33-5-day-upper- lower-specialty- advanced	SUCCESS	FAIL	30	5	advanced	Upper/Lower	p31-5-day-upper- priority-iranian- advanced
RECOVERY_SPACING_I NVALID	p33-5-day-upper- lower-specialty- advanced	SUCCESS	FAIL	45	5	advanced	Upper/Lower	p31-5-day-upper- priority-iranian- advanced
RECOVERY_SPACING_I NVALID	p33-5-day-upper- lower-specialty- advanced	SUCCESS	FAIL	60	5	advanced	Upper/Lower	p31-5-day-upper- priority-iranian- advanced
RECOVERY_SPACING_I NVALID	p33-5-day-upper- lower-specialty- advanced	SUCCESS	FAIL	75	5	advanced	Upper/Lower	p31-5-day-upper- priority-iranian- advanced
SESSION_DURATION_U NDER_TARGET	p33-5-day-upper- lower-specialty- advanced	SUCCESS	FAIL	90	5	advanced	Upper/Lower	p31-5-day-upper- priority-iranian- advanced
–	p26-5-day-classic- body-part- intermediate	SUCCESS	PASS	30	5	intermediate	Upper/Lower	p32-5-day-upper- lower-specialty- intermediate
–	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	PASS	45	5	intermediate	Upper/Lower	p32-5-day-upper- lower-specialty- intermediate
–	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	PASS	60	5	intermediate	Upper/Lower	p32-5-day-upper- lower-specialty- intermediate
–	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	PASS	75	5	intermediate	Upper/Lower	p32-5-day-upper- lower-specialty- intermediate
–	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	PASS	90	5	intermediate	Upper/Lower	p32-5-day-upper- lower-specialty- intermediate
–	p33-5-day-upper- lower-specialty- advanced	SUCCESS	PASS	30	5	advanced	Upper/Lower	p33-5-day-upper- lower-specialty- advanced

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
–	p33-5-day-upper-lower-specialty-advanced	SUCCESS	PASS	45	5	advanced	Upper/Lower	p33-5-day-upper-lower-specialty-advanced
–	p33-5-day-upper-lower-specialty-advanced	SUCCESS	PASS	60	5	advanced	Upper/Lower	p33-5-day-upper-lower-specialty-advanced
–	p33-5-day-upper-lower-specialty-advanced	SUCCESS	PASS	75	5	advanced	Upper/Lower	p33-5-day-upper-lower-specialty-advanced
–	p33-5-day-upper-lower-specialty-advanced	SUCCESS	PASS	90	5	advanced	Upper/Lower	p33-5-day-upper-lower-specialty-advanced
RECOVERY_SPACING_INVALID	p26-5-day-classic-body-part-intermediate	SUCCESS	FAIL	30	5	intermediate	Specialization	p34-5-day-fst7-arms-priority-intermediate
RECOVERY_SPACING_INVALID	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	45	5	intermediate	Specialization	p34-5-day-fst7-arms-priority-intermediate
RECOVERY_SPACING_INVALID	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	60	5	intermediate	Specialization	p34-5-day-fst7-arms-priority-intermediate
RECOVERY_SPACING_INVALID	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	75	5	intermediate	Specialization	p34-5-day-fst7-arms-priority-intermediate
SESSION_DURATION_UNDER_TARGET	p32-5-day-upper-lower-specialty-intermediate	SUCCESS	FAIL	90	5	intermediate	Specialization	p34-5-day-fst7-arms-priority-intermediate
RECOVERY_SPACING_INVALID	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	30	5	advanced	Specialization	p35-5-day-fst7-arms-priority-advanced
RECOVERY_SPACING_INVALID	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	45	5	advanced	Specialization	p35-5-day-fst7-arms-priority-advanced
SESSION_DURATION_UNDER_TARGET	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	60	5	advanced	Specialization	p35-5-day-fst7-arms-priority-advanced
SESSION_DURATION_UNDER_TARGET	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	75	5	advanced	Specialization	p35-5-day-fst7-arms-priority-advanced
SESSION_DURATION_UNDER_TARGET	p33-5-day-upper-lower-specialty-advanced	SUCCESS	FAIL	90	5	advanced	Specialization	p35-5-day-fst7-arms-priority-advanced

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
RECOVERY_SPACING_I NVALID	p26-5-day-classic- body-part- intermediate	SUCCESS	FAIL	30	5	intermediate	Body-Part	p36-5-day- professional- compound- intermediate
RECOVERY_SPACING_I NVALID	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	FAIL	45	5	intermediate	Body-Part	p36-5-day- professional- compound- intermediate
RECOVERY_SPACING_I NVALID	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	FAIL	60	5	intermediate	Body-Part	p36-5-day- professional- compound- intermediate
SESSION_DURATION_U NDER_TARGET	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	FAIL	75	5	intermediate	Body-Part	p36-5-day- professional- compound- intermediate
SESSION_DURATION_U NDER_TARGET	p32-5-day-upper- lower-specialty- intermediate	SUCCESS	FAIL	90	5	intermediate	Body-Part	p36-5-day- professional- compound- intermediate
RECOVERY_SPACING_I NVALID	p33-5-day-upper- lower-specialty- advanced	SUCCESS	FAIL	30	5	advanced	Body-Part	p37-5-day- professional- compound-advanced
RECOVERY_SPACING_I NVALID	p33-5-day-upper- lower-specialty- advanced	SUCCESS	FAIL	45	5	advanced	Body-Part	p37-5-day- professional- compound-advanced
SESSION_DURATION_U NDER_TARGET	p33-5-day-upper- lower-specialty- advanced	SUCCESS	FAIL	60	5	advanced	Body-Part	p37-5-day- professional- compound-advanced
SESSION_DURATION_U NDER_TARGET	p33-5-day-upper- lower-specialty- advanced	SUCCESS	FAIL	75	5	advanced	Body-Part	p37-5-day- professional- compound-advanced
SESSION_DURATION_U NDER_TARGET	p33-5-day-upper- lower-specialty- advanced	SUCCESS	FAIL	90	5	advanced	Body-Part	p37-5-day- professional- compound-advanced
RECOVERY_SPACING_I NVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	30	6	intermediate	PPLx2	p38-6-day-ppl-ab- intermediate
RECOVERY_SPACING_I NVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	45	6	intermediate	PPLx2	p38-6-day-ppl-ab- intermediate
RECOVERY_SPACING_I NVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	60	6	intermediate	PPLx2	p38-6-day-ppl-ab- intermediate

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
RECOVERY_SPACING_INVALID	p40-6-day-upper-lower-x3-intermediate	SUCCESS	FAIL	75	6	intermediate	PPLx2	p38-6-day-ppl-ab-intermediate
SESSION_DURATION_UPPER_TARGET	dynamic:None	FAILURE	FAIL	90	6	intermediate	PPLx2	p38-6-day-ppl-ab-intermediate
–	p39-6-day-ppl-ab-advanced	SUCCESS	PASS	30	6	advanced	PPLx2	p39-6-day-ppl-ab-advanced
RECOVERY_SPACING_INVALID	p41-6-day-upper-lower-x3-advanced	SUCCESS	FAIL	45	6	advanced	PPLx2	p39-6-day-ppl-ab-advanced
RECOVERY_SPACING_INVALID	p41-6-day-upper-lower-x3-advanced	SUCCESS	FAIL	60	6	advanced	PPLx2	p39-6-day-ppl-ab-advanced
RECOVERY_SPACING_INVALID	p41-6-day-upper-lower-x3-advanced	SUCCESS	FAIL	75	6	advanced	PPLx2	p39-6-day-ppl-ab-advanced
SESSION_DURATION_UPPER_TARGET	dynamic:upper_lower_x3	SUCCESS	FAIL	90	6	advanced	PPLx2	p39-6-day-ppl-ab-advanced
–	p40-6-day-upper-lower-x3-intermediate	SUCCESS	PASS	30	6	intermediate	Upper/Lower	p40-6-day-upper-lower-x3-intermediate
–	p40-6-day-upper-lower-x3-intermediate	SUCCESS	PASS	45	6	intermediate	Upper/Lower	p40-6-day-upper-lower-x3-intermediate
–	p40-6-day-upper-lower-x3-intermediate	SUCCESS	PASS	60	6	intermediate	Upper/Lower	p40-6-day-upper-lower-x3-intermediate
–	p40-6-day-upper-lower-x3-intermediate	SUCCESS	PASS	75	6	intermediate	Upper/Lower	p40-6-day-upper-lower-x3-intermediate
SESSION_DURATION_UPPER_TARGET	dynamic:None	FAILURE	FAIL	90	6	intermediate	Upper/Lower	p40-6-day-upper-lower-x3-intermediate
–	p39-6-day-ppl-ab-advanced	SUCCESS	PASS	30	6	advanced	Upper/Lower	p41-6-day-upper-lower-x3-advanced
–	p41-6-day-upper-lower-x3-advanced	SUCCESS	PASS	45	6	advanced	Upper/Lower	p41-6-day-upper-lower-x3-advanced
–	p41-6-day-upper-lower-x3-advanced	SUCCESS	PASS	60	6	advanced	Upper/Lower	p41-6-day-upper-lower-x3-advanced
–	p41-6-day-upper-lower-x3-advanced	SUCCESS	PASS	75	6	advanced	Upper/Lower	p41-6-day-upper-lower-x3-advanced
SESSION_DURATION_UPPER_TARGET	dynamic:upper_lower_x3	SUCCESS	FAIL	90	6	advanced	Upper/Lower	p41-6-day-upper-lower-x3-advanced

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
RECOVERY_SPACING_I NVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	30	6	intermediate	PPL	p42-6-day-fitclub- hybrid- intermediate
RECOVERY_SPACING_I NVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	45	6	intermediate	PPL	p42-6-day-fitclub- hybrid- intermediate
RECOVERY_SPACING_I NVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	60	6	intermediate	PPL	p42-6-day-fitclub- hybrid- intermediate
SESSION_DURATION_U NDER_TARGET	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	75	6	intermediate	PPL	p42-6-day-fitclub- hybrid- intermediate
SESSION_DURATION_U NDER_TARGET	dynamic:None	FAILURE	FAIL	90	6	intermediate	PPL	p42-6-day-fitclub- hybrid- intermediate
RECOVERY_SPACING_I NVALID	p39-6-day-ppl-ab- advanced	SUCCESS	FAIL	30	6	advanced	PPL	p43-6-day-fitclub- hybrid-advanced
RECOVERY_SPACING_I NVALID	p41-6-day-upper- lower-x3-advanced	SUCCESS	FAIL	45	6	advanced	PPL	p43-6-day-fitclub- hybrid-advanced
RECOVERY_SPACING_I NVALID	p41-6-day-upper- lower-x3-advanced	SUCCESS	FAIL	60	6	advanced	PPL	p43-6-day-fitclub- hybrid-advanced
SESSION_DURATION_U NDER_TARGET	p41-6-day-upper- lower-x3-advanced	SUCCESS	FAIL	75	6	advanced	PPL	p43-6-day-fitclub- hybrid-advanced
SESSION_DURATION_U NDER_TARGET	dynamic:upper_lowe r_x3	SUCCESS	FAIL	90	6	advanced	PPL	p43-6-day-fitclub- hybrid-advanced
RECOVERY_SPACING_I NVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	30	6	intermediate	Arnold-style	p44-6-day-arnold- split-intermediate
RECOVERY_SPACING_I NVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	45	6	intermediate	Arnold-style	p44-6-day-arnold- split-intermediate
RECOVERY_SPACING_I NVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	60	6	intermediate	Arnold-style	p44-6-day-arnold- split-intermediate
RECOVERY_SPACING_I NVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	75	6	intermediate	Arnold-style	p44-6-day-arnold- split-intermediate
SESSION_DURATION_U NDER_TARGET	dynamic:None	FAILURE	FAIL	90	6	intermediate	Arnold-style	p44-6-day-arnold- split-intermediate
RECOVERY_SPACING_I NVALID	p39-6-day-ppl-ab- advanced	SUCCESS	FAIL	30	6	advanced	Arnold-style	p45-6-day-arnold- split-advanced

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
RECOVERY_SPACING_I INVALID	p41-6-day-upper- lower-x3-advanced	SUCCESS	FAIL	45	6	advanced	Arnold-style	p45-6-day-arnold- split-advanced
RECOVERY_SPACING_I INVALID	p41-6-day-upper- lower-x3-advanced	SUCCESS	FAIL	60	6	advanced	Arnold-style	p45-6-day-arnold- split-advanced
SESSION_DURATION_U NDER_TARGET	p41-6-day-upper- lower-x3-advanced	SUCCESS	FAIL	75	6	advanced	Arnold-style	p45-6-day-arnold- split-advanced
SESSION_DURATION_U NDER_TARGET	dynamic:upper_lowe r_x3	SUCCESS	FAIL	90	6	advanced	Arnold-style	p45-6-day-arnold- split-advanced
RECOVERY_SPACING_I INVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	30	6	intermediate	Body-Part	p46-6-day-classic- body-part- intermediate
RECOVERY_SPACING_I INVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	45	6	intermediate	Body-Part	p46-6-day-classic- body-part- intermediate
SESSION_DURATION_U NDER_TARGET	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	60	6	intermediate	Body-Part	p46-6-day-classic- body-part- intermediate
SESSION_DURATION_U NDER_TARGET	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	75	6	intermediate	Body-Part	p46-6-day-classic- body-part- intermediate
SESSION_DURATION_U NDER_TARGET	dynamic:None	FAILURE	FAIL	90	6	intermediate	Body-Part	p46-6-day-classic- body-part- intermediate
RECOVERY_SPACING_I INVALID	p39-6-day-ppl-ab- advanced	SUCCESS	FAIL	30	6	advanced	Body-Part	p47-6-day-classic- body-part-advanced
RECOVERY_SPACING_I INVALID	p41-6-day-upper- lower-x3-advanced	SUCCESS	FAIL	45	6	advanced	Body-Part	p47-6-day-classic- body-part-advanced
SESSION_DURATION_U NDER_TARGET	p41-6-day-upper- lower-x3-advanced	SUCCESS	FAIL	60	6	advanced	Body-Part	p47-6-day-classic- body-part-advanced
SESSION_DURATION_U NDER_TARGET	p41-6-day-upper- lower-x3-advanced	SUCCESS	FAIL	75	6	advanced	Body-Part	p47-6-day-classic- body-part-advanced
SESSION_DURATION_U NDER_TARGET	dynamic:upper_lowe r_x3	SUCCESS	FAIL	90	6	advanced	Body-Part	p47-6-day-classic- body-part-advanced
RECOVERY_SPACING_I INVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	30	6	intermediate	Body-Part	p48-6-day-ronnie- double-exposure- intermediate
RECOVERY_SPACING_I INVALID	p40-6-day-upper- lower-x3- intermediate	SUCCESS	FAIL	45	6	intermediate	Body-Part	p48-6-day-ronnie- double-exposure- intermediate

Primary failure	Final selected	Competition	Forced	Duration	Days	Level	Topology	Template
RECOVERY_SPACING_INVALID	p40-6-day-upper-lower-x3-intermediate	SUCCESS	FAIL	60	6	intermediate	Body-Part	p48-6-day-ronnie-double-exposure-intermediate
RECOVERY_SPACING_INVALID	p40-6-day-upper-lower-x3-intermediate	SUCCESS	FAIL	75	6	intermediate	Body-Part	p48-6-day-ronnie-double-exposure-intermediate
SESSION_DURATION_UNDER_TARGET	dynamic:None	FAILURE	FAIL	90	6	intermediate	Body-Part	p48-6-day-ronnie-double-exposure-intermediate
RECOVERY_SPACING_INVALID	p39-6-day-ppl-ab-advanced	SUCCESS	FAIL	30	6	advanced	Body-Part	p49-6-day-ronnie-double-exposure-advanced
RECOVERY_SPACING_INVALID	p41-6-day-upper-lower-x3-advanced	SUCCESS	FAIL	45	6	advanced	Body-Part	p49-6-day-ronnie-double-exposure-advanced
RECOVERY_SPACING_INVALID	p41-6-day-upper-lower-x3-advanced	SUCCESS	FAIL	60	6	advanced	Body-Part	p49-6-day-ronnie-double-exposure-advanced
RECOVERY_SPACING_INVALID	p41-6-day-upper-lower-x3-advanced	SUCCESS	FAIL	75	6	advanced	Body-Part	p49-6-day-ronnie-double-exposure-advanced
SESSION_DURATION_UNDER_TARGET	dynamic:upper_lower_x3	SUCCESS	FAIL	90	6	advanced	Body-Part	p49-6-day-ronnie-double-exposure-advanced

Template Survival Summary .3

Success rate	FAIL	PASS	Tests	Topology	Template
80.0%	1	4	5	Upper/Lower	p16-4-day-upper-lower-upper-lower-intermediate
80.0%	1	4	5	Upper/Lower	p17-4-day-upper-lower-upper-lower-advanced
100.0%	0	5	5	Upper/Lower	p19-4-day-3-upper-1-lower-intermediate
100.0%	0	5	5	Upper/Lower	p20-4-day-3-upper-1-lower-advanced
60.0%	2	3	5	Upper/Lower	p22-4-day-3-lower-1-upper-intermediate
40.0%	3	2	5	Upper/Lower	p23-4-day-3-lower-1-upper-advanced
100.0%	0	5	5	PPL	p24-4-day-push-pull-quads-posterior-intermediate
100.0%	0	5	5	PPL	p25-4-day-push-pull-quads-posterior-advanced
100.0%	0	5	5	Body-Part	p50-4-day-iranmuscle-intermediate
100.0%	0	5	5	Body-Part	p51-4-day-gymextreme-advanced
80.0%	1	4	5	Arnold-style	p52-4-day-arnoldsho-advanced
40.0%	3	2	5	Body-Part	p53-4-day-aloplay-intermediate
20.0%	4	1	5	Body-Part	p26-5-day-classic-body-part-intermediate
0.0%	5	0	5	Body-Part	p27-5-day-classic-body-part-advanced
0.0%	5	0	5	Body-Part	p28-5-day-split-weak-point-intermediate
0.0%	5	0	5	Body-Part	p29-5-day-split-weak-point-advanced

Success rate	FAIL	PASS	Tests	Topology	Template
0.0%	5	0	5	Upper/Lower	p30-5-day-upper-priority-iranian-intermediate
0.0%	5	0	5	Upper/Lower	p31-5-day-upper-priority-iranian-advanced
100.0%	0	5	5	Upper/Lower	p32-5-day-upper-lower-specialty-intermediate
100.0%	0	5	5	Upper/Lower	p33-5-day-upper-lower-specialty-advanced
0.0%	5	0	5	Specialization	p34-5-day-fst7-arms-priority-intermediate
0.0%	5	0	5	Specialization	p35-5-day-fst7-arms-priority-advanced
0.0%	5	0	5	Body-Part	p36-5-day-professional-compound-intermediate
0.0%	5	0	5	Body-Part	p37-5-day-professional-compound-advanced
0.0%	5	0	5	PPLx2	p38-6-day-ppl-ab-intermediate
20.0%	4	1	5	PPLx2	p39-6-day-ppl-ab-advanced
80.0%	1	4	5	Upper/Lower	p40-6-day-upper-lower-x3-intermediate
80.0%	1	4	5	Upper/Lower	p41-6-day-upper-lower-x3-advanced
0.0%	5	0	5	PPL	p42-6-day-fitclub-hybrid-intermediate
0.0%	5	0	5	PPL	p43-6-day-fitclub-hybrid-advanced
0.0%	5	0	5	Arnold-style	p44-6-day-arnold-split-intermediate
0.0%	5	0	5	Arnold-style	p45-6-day-arnold-split-advanced
0.0%	5	0	5	Body-Part	p46-6-day-classic-body-part-intermediate
0.0%	5	0	5	Body-Part	p47-6-day-classic-body-part-advanced
0.0%	5	0	5	Body-Part	p48-6-day-ronnie-double-exposure-intermediate

Success rate	FAIL	PASS	Tests	Topology	Template
0.0%	5	0	5	Body-Part	p49-6-day-ronnie-double-exposure-advanced

Detailed Failures .4

هر بلوک یک Forced failure واقعی است. جدول روزانه، وضعیت بعد از session building و آخرین state پیش از reject را مقایسه می‌کند. نسبت Repair به شکل success/call نمایش داده شده است.

1. min 30 — intermediate — p16-4-day-upper-lower-upper-lower-intermediate

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TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED :Root cause | template_construction :Failure stage
INITIAL_TEMPLATE_REJECTED_UNFILLABLE :Reason codes
TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED
TEMPLATE_DAY:1
TEMPLATE_MAIN_COUNT_OUT_OF_RANGE
validate_program() :: app/workouts/program_engine/validation.py :Relevant
template_construction:failed[TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED,TEMPLATE_DAY:1,TEMPLATE_MAIN_COUNT_OUT_OF_RANGE] :Stage chain
Session ساخته نشد؛ failure در construction/hard eligibility رخ داد. Catalog day slot counts: 6
5
6
5
```

2. min 30 — advanced — p17-4-day-upper-lower-upper-lower-advanced

```
TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED :Root cause | template_construction :Failure stage
INITIAL_TEMPLATE_REJECTED_UNFILLABLE :Reason codes
TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED
TEMPLATE_DAY:1
TEMPLATE_MAIN_COUNT_OUT_OF_RANGE
validate_program() :: app/workouts/program_engine/validation.py :Relevant
template_construction:failed[TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED,TEMPLATE_DAY:1,TEMPLATE_MAIN_COUNT_OUT_OF_RANGE] :Stage chain
Session ساخته نشد؛ failure در construction/hard eligibility رخ داد. Catalog day slot counts: 6
5
6
5
```

min 30 — intermediate — p22-4-day-3-lower-1-upper-intermediate .3

TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED :Root cause | template_construction :Failure stage
INITIAL_TEMPLATE_REJECTED_UNFILLABLE :Reason codes
TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED
TEMPLATE_DAY:2
TEMPLATE_MAIN_COUNT_OUT_OF_RANGE
validate_program() :: app/workouts/program_engine/validation.py :Relevant
template_construction:failed[TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED,TEMPLATE_DAY:2,TEMPLATE_MAIN_COUNT_OUT_OF_RANGE] :Stage chain
4:Catalog day slot counts رخ داد. Session ساخته نشد؛ failure در construction/hard eligibility
8
4
4

min 90 — intermediate — p22-4-day-3-lower-1-upper-intermediate .4

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage
SESSION_DURATION_UNDER_TARGET :Reason codes
SESSION_DURATION_TARGET_UNSATISFIED
SESSION_DURATION_CONSTRAINT_UNEXPLAINED
repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant
template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINED_BY_USEFUL_WORKLOAD]
recovery_repair:completed[RECOVERY_WEEKDAYS_REARRANGED_FOR_DIRECT_MUSCLE_OVERLAP,RECOVERY_WEEKDAYS_REARRANGED_FOR_EXPOSURE_LOAD]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINED_BY_USEFUL_WORKLOAD]
validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED]
final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:735 no_useful_candidate:170 exercise_set_cap:35 duplicate:35 near_equivalent:25 weekly_hard_volume_cap:16 acceptable_volume_preference:8	5/5	8/13	82	39	10	55	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:46 no_useful_candidate:34 exercise_set_cap:8 duplicate:8 near_equivalent:3	1/1	0/1	81	35	9	75	32	8	template_reference_2	2
UNDER_TARGET	wrong_template_muscle:800 no_useful_candidate:170 weekly_hard_volume_cap:110 acceptable_volume_preference:50 duplicate:29 near_equivalent:25 exercise_set_cap:11	1/5	4/7	50	22	6	55	24	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:1106 no_useful_candidate:238 weekly_hard_volume_cap:186 acceptable_volume_preference:82 duplicate:50 exercise_set_cap:24 near_equivalent:7	3/7	6/11	65	30	8	54	24	8	template_reference_4	4

min 30 — advanced — p23-4-day-3-lower-1-upper-advanced .5

TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED :Root cause | template_construction :Failure stage
INITIAL_TEMPLATE_REJECTED_UNFILLABLE :Reason codes
TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED
TEMPLATE_DAY:2
TEMPLATE_MAIN_COUNT_OUT_OF_RANGE
validate_program() :: app/workouts/program_engine/validation.py :Relevant
template_construction:failed[TEMPLATE_SESSION_EXERCISE_COUNT_UNSATISFIED,TEMPLATE_DAY:2,TEMPLATE_MAIN_COUNT_OUT_OF_RANGE] :Stage chain
4 :Catalog day slot counts رخ داد. construction/hard eligibility در failure ساخته نشد؛
8
4
4

min 60 — advanced — p23-4-day-3-lower-1-upper-advanced .6

SESSION_DURATION_TARGET_UNSATISFIED :Root cause | final_quality_gate :Failure stage
SESSION_DURATION_EXCEEDED :Reason codes
SESSION_DURATION_OVER_TARGET

```
SESSION_DURATION_TARGET_UNSATISFIED
SESSION_DURATION_CONSTRAINT_UNEXPLAINED
repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant
template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAYS_REARRANGED_FOR_DIRECT_MUSCLE_OVERLAP,RECOVERY_WEEKDAYS_REARRANGED_FOR_EXPOSURE_LOAD]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_OVERFILLED,SESSION_DURATION_REPAIR_APPLIED]
validation:failed[SESSION_DURATION_EXCEEDED,SESSION_DURATION_OVER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED]
final_quality_gate:rejected[SESSION_DURATION_EXCEEDED,SESSION_DURATION_OVER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]
```

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	4/4	52	22	6	54	24	8	template_ref erence_1	1
OVER_TARGET	—	0/0	0/0	71	30	8	55	24	8	template_ref erence_2	2
DAY_LIMITS_0 K	wrong_templa te_muscle: 164 no_useful_ca ndidate:34 near_equiv alent:7 duplicate:5 exercise_set _cap:4 weekly_hard volume_cap:1	1/1	4/5	51	22	6	54	24	8	template_ref erence_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:328 no_useful_candidate:68 weekly_hard_volume_cap:28 duplicate:11 acceptable_volume_preference:8 exercise_set_cap:7 near_equivalent:2	2/2	4/6	56	25	7	54	24	8	template_reference_4	4

min 90 — advanced — p23-4-day-3-lower-1-upper-advanced .7

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REduced_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINED_BY_USEFUL_WORKLOAD]

recovery_repair:completed[RECOVERY_WEEKDAYS_REARRANGED_FOR_DIRECT_MUSCLE_OVERLAP,RECOVERY_WEEKDAYS_REARRANGED_FOR_EXPOSURE_LOAD]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINED_BY_USEFUL_WORKLOAD]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:453 no_useful_candidate:102 near_equivalent:36 exercise_set_cap:21 duplicate:21 weekly_hard_volume_cap:12 acceptable_volume_preference:6	3/3	8/11	81	36	9	58	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:49 no_useful_candidate:34 exercise_set_cap:8 duplicate:8 near_equivalent:3	1/1	0/1	82	35	9	75	32	8	template_reference_2	2
UNDER_TARGET	wrong_template_muscle:1148 no_useful_candidate:238 weekly_hard_volume_cap:130 acceptable_volume_preference:57 near_equivalent:53 duplicate:50 exercise_set_cap:24	3/7	6/11	67	30	8	54	24	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:1148 no_useful_candidate:238 weekly_hard_volume_cap:168 acceptable_volume_preference:73 duplicate:50 near_equivalent:31 exercise_set_cap:24	3/7	6/11	69	30	8	54	24	8	template_reference_4	4

min 90 — advanced — p52-4-day-arnoldsho-advanced .8

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REduced_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINED_BY_USEFUL_WORKLOAD]

recovery_repair:completed[RECOVERY_WEEKDAYS_REARRANGED_FOR_DIRECT_MUSCLE_OVERLAP,RECOVERY_WEEKDAYS_REARRANGED_FOR_EXPOSURE_LOAD]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINED_BY_USEFUL_WORKLOAD]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 476 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:6	4/4	4/8	82	35	9	64	28	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle: 357 no_useful_candidate:102 exercise_set_cap:21 duplicate:21 near_equivalent:6	3/3	5/8	82	35	9	47	21	6	template_reference_2	2
UNDER_TARGET	wrong_template_muscle: 1078 no_useful_candidate:238 weekly_hard_volume_cap: 212 acceptable_volume_preference:106 duplicate:50 exercise_set_cap:34 near_equivalent:7	3/7	8/13	67	32	8	55	25	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle: 302 no_useful_candidate:68 near_equivalent:22 exercise_set_cap:17 duplicate:17	2/2	1/3	85	39	10	71	32	8	template_reference_4	4

min 30 — intermediate — p53-4-day-aloplay-intermediate .9

REQUIRED_MOVEMENT_PATTERN_MISSING :Root cause | final_quality_gate :Failure stage

REQUIRED_MOVEMENT_PATTERN_MISSING :Reason codes

validate_program() :: app/workouts/program_engine/validation.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARGET_REDUCE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAYS_REARRANGED_FOR_DIRECT_MUSCLE_OVERLAP,RECOVERY_WEEKDAYS_REARRANGED_FOR_EXPOSURE_LOAD]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[REQUIRED_MOVEMENT_PATTERN_MISSING]

final_quality_gate:rejected[REQUIRED_MOVEMENT_PATTERN_MISSING]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0K	—	0/0	0/0	27	11	3	28	12	4	template_reference_1	1
DAY_LIMITS_0K	—	0/0	0/0	35	16	4	27	12	4	template_reference_2	2
DAY_LIMITS_0K	—	0/0	0/0	37	16	4	28	12	4	template_reference_3	3
DAY_LIMITS_0K	—	0/0	0/0	35	16	4	27	12	4	template_reference_4	4

min 45 — intermediate — p53-4-day-aloplay-intermediate .10

REQUIRED_MOVEMENT_PATTERN_MISSING :Root cause | final_quality_gate :Failure stage

REQUIRED_MOVEMENT_PATTERN_MISSING :Reason codes

validate_program() :: app/workouts/program_engine/validation.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARGET_REDUCE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]
recovery_repair:completed[RECOVERY_WEEKDAYS_REARRANGED_FOR_DIRECT_MUSCLE_OVERLAP,RECOVERY_WEEKDAYS_REARRANGED_FOR_EXPOSURE_LOAD]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[REQUIRED_MOVEMENT_PATTERN_MISSING]
final_quality_gate:rejected[REQUIRED_MOVEMENT_PATTERN_MISSING]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	46	20	6	41	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	43	20	6	39	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	45	20	6	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	1/1	35	16	5	39	18	6	template_ref erence_4	4

min 90 — intermediate — p53-4-day-aloplay-intermediate .11

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINED_BY_USEFUL_WORKLOAD]

recovery_repair:completed[RECOVERY_WEEKDAYS_REARRANGED_FOR_DIRECT_MUSCLE_OVERLAP,RECOVERY_WEEKDAYS_REARRANGED_FOR_EXPOSURE_LOAD]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINED_BY_USEFUL_WORKLOAD]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 342 no_useful_candidate:102 exercise_set_cap:21 duplicate:21 near_equivalent:8	3/3	5/8	80	35	9	64	28	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle: 294 no_useful_candidate:68 exercise_set_cap:17 duplicate:17 near_equivalent:10	2/2	5/7	81	39	10	59	28	8	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle: 342 no_useful_candidate:102 exercise_set_cap:21 duplicate:21 near_equivalent:12	3/3	6/9	83	35	9	45	20	6	template_reference_3	3
UNDER_TARGET	wrong_template_muscle: 1043 no_useful_candidate:234 weekly_hard_volume_cap: 216 acceptable_volume_preference:108 duplicate:64 exercise_set_cap:34	3/7	8/13	67	32	8	52	24	8	template_reference_4	4

min 45 — intermediate — p26-5-day-classic-body-part-intermediate .12

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	41	18	5	36	16	5	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	35	15	5	42	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	1/1	35	16	5	39	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	43	20	6	39	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	53	24	6	40	18	6	template_ref erence_5	5

min 60 — intermediate — p26-5-day-classic-body-part-intermediate .13

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:140 no_useful_candidate:34 duplicate:5 exercise_set_cap:4 weekly_hard_volume_cap:2 near_equivalent:1	1/1	1/2	51	22	6	41	18	5	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:110 no_useful_candidate:34 weekly_hard_volume_cap:10 duplicate:5 exercise_set_cap:3 near_equivalent:2	1/1	4/5	52	22	6	42	18	6	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:745 weekly_hard_volume_cap:238 no_useful_candidate:170 acceptable_volume_preference:110 duplicate:29 exercise_set_cap:8	1/5	3/6	45	21	6	53	24	8	template_reference_3	3
DAY_LIMITS_0K	—	0/0	4/4	51	24	6	53	24	8	template_reference_4	4
DAY_LIMITS_0K	—	0/0	0/0	61	28	8	52	24	8	template_reference_5	5

min 75 — intermediate — p26-5-day-classic-body-part-intermediate .14

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:420 no_useful_candidate:102 duplicate:18 exercise_set_cap:15 near_equivalent:7 weekly_hard_volume_cap:6	3/3	3/6	70	30	8	41	18	5	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:330 no_useful_candidate:102 duplicate:18 weekly_hard_volume_cap:16 exercise_set_cap:12 near_equivalent:6	3/3	5/8	70	29	8	42	18	6	template_reference_2	2
UNDER_TARGET	wrong_template_muscle:745 weekly_hard_volume_cap:238 no_useful_candidate:170 acceptable_volume_preferance:110 duplicate:29 exercise_set_cap:8	1/5	3/6	45	21	6	53	24	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:424 no_useful_candidate:136 weekly_hard_volume_cap:80 acceptable_volume_preference:40 exercise_set_cap:26 duplicate:26 near_equivalent:18	4/4	4/8	70	35	9	61	28	8	template_reference_4	4
DAY_LIMITS_0 K	—	0/0	2/2	65	30	8	58	27	8	template_reference_5	5

min 90 — intermediate — p26-5-day-classic-body-part-intermediate .15

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:560 no_useful_candidate:136 duplicate:26 exercise_set_cap:22 near_equivalent:10 weekly_hard_volume_cap:8	4/4	4/8	80	34	9	41	18	5	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:550 no_useful_candidate:170 session_volume_cap:43 duplicate:35 exercise_set_cap:24 weekly_hard_volume_cap:23 near_equivalent:10 acceptable_volume_preference:1	5/5	6/11	85	36	10	42	18	6	template_reference_2	2
UNDER_TARGET	wrong_template_muscle:745 weekly_hard_volume_cap:237 no_useful_candidate:170 acceptable_volume_preference:110 duplicate:29 exercise_set_cap:6	1/5	2/5	43	20	6	53	24	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:636 weekly_hard_volume_cap:251 no_useful_candidate:204 acceptable_volume_preference:120 duplicate:45 exercise_set_cap:39 near_equivalent:28	6/6	5/11	84	42	11	61	28	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:294 no_useful_candidate:68 near_equivalent:18 exercise_set_cap:17 duplicate:17	2/2	5/7	83	39	10	61	28	8	template_reference_5	5

min 30 — advanced — p27-5-day-classic-body-part-advanced .16

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage
RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant
template_construction:succeeded :Stage chain

session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SAFE_TEMPLATE_SUPERSET_PRESERVED,SESSION_DURATION_REPAIR_APPLIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	33	14	4	28	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	29	12	4	29	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	23	10	3	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	28	16	4	24	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	39	16	4	29	12	4	template_ref erence_5	5

min 45 — advanced — p27-5-day-classic-body-part-advanced .17

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SAFE_TEMPLATE_SUPERSET_PRESERVED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	43	18	5	35	15	5	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	36	15	5	36	15	5	template_ref erence_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	40	18	6	40	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	40	20	6	39	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	55	23	6	42	18	6	template_ref erence_5	5

min 60 — advanced — p27-5-day-classic-body-part-advanced .18

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SAFE_TEMPLATE_SUPERSET_PRESERVED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_templa te_muscle: 145 no_useful_ca ndidate:34 exercise_set _cap:5 duplicate:5 near_equivale nt:1	1/1	2/3	54	23	6	43	18	5	template_ref erence_1	1

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:142 no_useful_candidate:34 exercise_set_cap:5 duplicate:5	1/1	4/5	56	23	6	38	16	5	template_reference_2	2
DAY_LIMITS_0 K	weekly_hard_volume_cap:8	0/0	2/2	52	23	7	54	24	8	template_reference_3	3
DAY_LIMITS_0 K	—	0/0	0/0	54	26	8	53	24	8	template_reference_4	4
DAY_LIMITS_0 K	—	0/0	0/0	69	29	8	55	24	8	template_reference_5	5

min 75 — advanced — p27-5-day-classic-body-part-advanced .19

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,SAFE_TEMPLATE_SUPERSET_PRESERVED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:290 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:3	2/2	4/6	67	28	7	43	18	5	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:284 no_useful_candidate:68 exercise_set_cap:11 duplicate:11	2/2	5/7	66	27	7	38	16	5	template_reference_2	2
UNDER_TARGET	wrong_template_muscle:924 weekly_hard_volume_cap:234 no_useful_candidate:204 acceptable_volume_preference:108 duplicate:45 exercise_set_cap:19 near_equivalent:6	2/6	4/8	64	29	8	56	25	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:222 no_useful_candidate:68 duplicate:15 exercise_set_cap:13 near_equivalent:8 weekly_hard_volume_cap:5	2/2	3/5	69	34	9	61	28	8	template_reference_4	4
DAY_LIMITS_0 K	—	0/0	0/0	75	32	8	61	27	8	template_reference_5	5

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CO

NSTRAINED_BY_HARD_VOLUME_LIMITS,SAFE_TEMPLATE_SUPERSET_PRESERVED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0K	wrong_template_muscle:580 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:7	4/4	5/9	84	35	9	43	18	5	template_reference_1	1
UNDER_TARGET	wrong_template_muscle:852 no_useful_candidate:204 session_volume_cap:160 duplicate:39 exercise_set_cap:25	2/6	6/10	69	28	7	38	16	5	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:924 weekly_hard_volume_cap:234 no_useful_candidate:204 acceptable_volume_preference:108 duplicate:45 exercise_set_cap:19 near_equivalent:6	2/6	4/8	64	29	8	56	25	8	template_reference_3	3
DAY_LIMITS_0_K	wrong_template_muscle:444 no_useful_candidate:136 weekly_hard_volume_cap:95 acceptable_volume_preference:43 duplicate:34 exercise_set_cap:30 near_equivalent:16	4/4	5/9	85	42	11	61	28	8	template_reference_4	4
DAY_LIMITS_0_K	wrong_template_muscle:151 no_useful_candidate:34 near_equivalent:11 exercise_set_cap:8 duplicate:8	1/1	0/1	81	35	9	75	32	8	template_reference_5	5

min 30 — intermediate — p28-5-day-split-weak-point-intermediate .21

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	37	16	4	28	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	30	13	4	28	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	37	16	4	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	27	12	4	27	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	38	15	4	30	12	4	template_ref erence_5	5

min 45 — intermediate — p28-5-day-split-weak-point-intermediate .22

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	46	20	6	41	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	38	17	5	41	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	47	21	6	40	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	1/1	35	16	5	39	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	47	20	6	42	18	6	template_ref erence_5	5

min 60 — intermediate — p28-5-day-split-weak-point-intermediate .23

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage
RECOVERY_SPACING_INVALID :Reason codes
recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant
template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	4/4	51	22	6	55	24	8	template_ref erence_1	1
DAY_LIMITS_0 K	wrong_templa te_muscle: 114 no_useful_ca ndidate:34 exercise_set _cap:5 duplicate:5 near_equival ent:4	1/1	3/4	52	23	6	45	20	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	55	25	7	52	24	8	template_ref erence_3	3
DAY_LIMITS_0 K	wrong_templa te_muscle: 298 no_useful_ca ndidate:68 weekly_hard_ volume_cap: 16 duplicate:15 exercise_set _cap:7 acceptable_v olume_prefer ence:4	2/2	4/6	53	25	7	53	24	8	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	57	25	8	54	24	8	template_ref erence_5	5

min 75 — intermediate — p28-5-day-split-weak-point-intermediate .24

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage
SESSION_DURATION_UNDER_TARGET :Reason codes
SESSION_DURATION_TARGET_UNSATISFIED
RECOVERY_SPACING_INVALID
SESSION_DURATION_CONSTRAINT_UNEXPLAINED
repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant
template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:342 no_useful_candidate:102 exercise_set_cap:18 duplicate:18 near_equivalent:5	3/3	5/8	70	31	8	59	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:228 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:8	2/2	5/7	65	28	7	45	20	6	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle:147 no_useful_candidate:34 near_equivalent:9 exercise_set_cap:7 duplicate:7	1/1	3/4	67	31	8	58	27	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:1043 weekly_hard_volume_cap:234 no_useful_candidate:234 acceptable_volume_preference:108 duplicate:64 exercise_set_cap:22	3/7	5/10	61	29	8	53	24	8	template_reference_4	4
DAY_LIMITS_OK	weekly_hard_volume_cap:4	0/0	4/4	66	29	8	54	24	8	template_reference_5	5

min 90 — intermediate — p28-5-day-split-weak-point-intermediate .25

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 456 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:8	4/4	6/10	80	35	9	59	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle: 456 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:16	4/4	6/10	82	35	9	45	20	6	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle: 441 no_useful_candidate:102 near_equivalent:27 exercise_set_cap:24 duplicate:24	3/3	5/8	83	39	10	61	28	8	template_reference_3	3
UNDER_TARGET	wrong_template_muscle: 1043 weekly_hard_volume_cap: 234 no_useful_candidate:234 acceptable_volume_preference:108 duplicate:64 exercise_set_cap:22	3/7	5/10	61	29	8	53	24	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	weekly_hard_volume_cap:195 acceptable_volume_preference:88 no_useful_candidate:68 wrong_template_muscle:22 duplicate:17 exercise_set_cap:11 near_equivalent:3	2/2	5/7	80	36	10	57	25	8	template_reference_5	5

min 30 — advanced — p29-5-day-split-weak-point-advanced .26

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	35	15	4	28	12	4	template_reference_1	1
DAY_LIMITS_0 K	—	0/0	0/0	30	13	4	28	12	4	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	39	16	4	29	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	28	12	4	28	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	38	15	4	30	12	4	template_ref erence_5	5

min 45 — advanced — p29-5-day-split-weak-point-advanced .27

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	46	20	6	41	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	38	17	5	41	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	52	23	6	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	1/1	36	16	5	40	18	6	template_ref erence_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	54	22	6	43	18	6	template_ref erence_5	5

min 60 — advanced — p29-5-day-split-weak-point-advanced .28

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	2/2	51	22	6	55	24	8	template_ref erence_1	1
DAY_LIMITS_0 K	wrong_templa te_muscle: 119 no_useful_ca ndidate:34 exercise_set _cap:5 duplicate:5 near_equiv alent:3	1/1	3/4	52	23	6	45	20	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	63	28	8	54	24	8	template_ref erence_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:154 no_useful_candidate:34 duplicate:6 exercise_set_cap:5 weekly_hard_volume_cap:2 acceptable_volume_preference:1 near_equivalent:1	1/1	5/6	51	23	6	54	24	8	template_reference_4	4
DAY_LIMITS_0 K	—	0/0	0/0	65	27	8	56	24	8	template_reference_5	5

min 75 — advanced — p29-5-day-split-weak-point-advanced .29

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:238 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:2	2/2	3/5	65	28	7	59	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:238 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:6	2/2	4/6	65	28	7	47	21	6	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	1/1	66	29	8	60	27	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:462 no_useful_candidate:102 duplicate:21 exercise_set_cap:18 weekly_hard_volume_cap:14 acceptable_volume_preference:7 near_equivalent:3	3/3	6/9	67	31	8	56	25	8	template_reference_4	4
DAY_LIMITS_0 K	weekly_hard_volume_cap:1	0/0	1/1	67	28	8	56	24	8	template_reference_5	5

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0K	wrong_template_muscle:476 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:6	4/4	4/8	82	35	9	64	28	8	template_reference_1	1

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 476 no_useful_candidate:136 weekly_hard_volume_cap: 82 acceptable_volume_preference:41 exercise_set_cap:26 duplicate:26 near_equivalent:12	4/4	5/9	81	35	9	47	21	6	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle: 151 no_useful_candidate:34 near_equivalent:11 exercise_set_cap:8 duplicate:8	1/1	4/5	81	36	9	66	29	8	template_reference_3	3
UNDER_TARGET	wrong_template_muscle: 1078 no_useful_candidate:236 weekly_hard_volume_cap: 224 acceptable_volume_preference:111 duplicate:57 exercise_set_cap:32 near_equivalent:7	3/7	6/11	67	31	8	56	25	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	weekly_hard_volume_cap:235 acceptable_volume_preference:110 no_useful_candidate:68 wrong_template_muscle:22 near_equivalent:18 duplicate:17 exercise_set_cap:11	2/2	3/5	84	36	10	64	27	8	template_reference_5	5

min 30 — intermediate — p30-5-day-upper-priority-iranian-intermediate .31

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

REQUIRED_MOVEMENT_PATTERN_MISSING

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARGET_REduced]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID,REQUIRED_MOVEMENT_PATTERN_MISSING]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID,REQUIRED_MOVEMENT_PATTERN_MISSING]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	28	12	4	28	12	4	template_reference_1	1

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	37	16	4	28	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	35	16	4	27	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	23	10	3	28	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	35	15	4	28	12	4	template_ref erence_5	5

min 45 — intermediate — p30-5-day-upper-priority-iranian-intermediate .32

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

REQUIRED_MOVEMENT_PATTERN_MISSING

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARG
ET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID,REQUIRED_MOVEMENT_PATTERN_MISSING]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID,REQUIRED_MOVEMENT_PATTERN_MISSING]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	1/1	36	16	5	40	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	1/1	35	16	5	40	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	43	20	6	39	18	6	template_ref erence_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	1/1	36	16	5	40	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	wrong_templa te_muscle:46 no_useful_ca ndidate:33 duplicate:5 near_equival ent:1	1/1	2/2	42	19	5	40	18	6	template_ref erence_5	5

min 60 — intermediate — p30-5-day-upper-priority-iranian-intermediate .33

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

REQUIRED_MOVEMENT_PATTERN_MISSING

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARG
ET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID,REQUIRED_MOVEMENT_PATTERN_MISSING]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID,REQUIRED_MOVEMENT_PATTERN_MISSING]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_templa te_muscle:46 no_useful_ca ndidate:34 exercise_set _cap:5 duplicate:5 near_equival ent:2	1/1	5/6	51	23	6	52	24	8	template_ref erence_1	1

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:92 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:2	2/2	6/8	57	27	7	52	24	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	0/0	59	28	8	51	24	8	template_reference_3	3
DAY_LIMITS_0 K	weekly_hard_volume_cap:44 wrong_template_muscle:40 no_useful_candidate:34 acceptable_volume_preference:22 exercise_set_cap:5 duplicate:5 near_equivalent:1	1/1	5/6	51	24	6	52	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:46 no_useful_candidate:34 duplicate:6 exercise_set_cap:5 near_equivalent:2	1/1	3/4	50	23	6	52	24	8	template_reference_5	5

min 75 — intermediate — p30-5-day-upper-priority-iranian-intermediate .34

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

REQUIRED_MOVEMENT_PATTERN_MISSING

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID,REQUIRED_MOVEMENT_PATTERN_MISSING]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID,REQUIRED_MOVEMENT_PATTERN_MISSING]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:92 no_useful_candidate:68 exercise_set_cap:13 duplicate:13 near_equivalent:4	2/2	5/7	68	31	8	60	28	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:138 no_useful_candidate:102 exercise_set_cap:18 duplicate:18 near_equivalent:4	3/3	7/10	66	31	8	52	24	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	3/3	65	31	8	59	28	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	weekly_hard_volume_cap:132 wrong_template_muscle:120 no_useful_candidate:102 acceptable_volume_preference:66 exercise_set_cap:18 duplicate:18 near_equivalent:5	3/3	6/9	68	31	8	54	25	8	template_reference_4	4
DAY_LIMITS_0 K	weekly_hard_volume_cap:325 acceptable_volume_preference:162 wrong_template_muscle:138 no_useful_candidate:102 duplicate:21 exercise_set_cap:17 near_equivalent:8	3/3	4/7	67	30	8	54	25	8	template_reference_5	5

min 90 — intermediate — p30-5-day-upper-priority-iranian-intermediate .35

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

REQUIRED_MOVEMENT_PATTERN_MISSING

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,REQUIRED_MOVEMENT_PATTERN_MISSING]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,REQUIRED_MOVEMENT_PATTERN_MISSING,SESSION_DURATION_CONSTR
AINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:138 no_useful_candidate:102 exercise_set_cap:21 duplicate:21 near_equivalent:7	3/3	7/10	81	36	9	60	28	8	template_reference_1	1
DAY_LIMITS_0 K	weekly_hard_volume_cap:262 wrong_template_muscle:230 no_useful_candidate:170 acceptable_volume_preference:125 duplicate:35 exercise_set_cap:30 near_equivalent:8	5/5	8/13	81	38	10	52	24	8	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 294 no_useful_candidate:68 duplicate:19 exercise_set_cap:17 near_equivalent:10	2/2	5/7	81	39	10	59	28	8	template_reference_3	3
DAY_LIMITS_0 K	weekly_hard_volume_cap: 203 wrong_template_muscle: 160 no_useful_candidate:136 acceptable_volume_preference:96 duplicate:30 exercise_set_cap:25 near_equivalent:13	4/4	6/10	82	37	10	54	25	8	template_reference_4	4
UNDER_TARGET	weekly_hard_volume_cap: 1245 acceptable_volume_preference:615 wrong_template_muscle: 230 no_useful_candidate:168 duplicate:34 near_equivalent:10 exercise_set_cap:8	1/5	2/5	45	21	6	59	27	8	template_reference_5	5

min 30 — advanced — p31-5-day-upper-priority-iranian-advanced .36

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	30	13	4	28	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	27	12	4	27	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	39	16	4	29	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	29	12	4	29	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	29	12	4	29	12	4	template_ref erence_5	5

min 45 — advanced — p31-5-day-upper-priority-iranian-advanced .37

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	44	20	6	40	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	1/1	36	16	5	40	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	55	24	6	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	42	18	6	42	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	1/1	35	16	5	40	18	6	template_ref erence_5	5

min 60 — advanced — p31-5-day-upper-priority-iranian-advanced .38

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage
RECOVERY_SPACING_INVALID :Reason codes
recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant
template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:49 no_useful_candidate:34 exercise_set_cap:5 duplicate:5	1/1	5/6	54	23	6	54	24	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:49 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:2	1/1	5/6	52	23	6	52	24	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	0/0	70	30	8	55	24	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:49 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:2	1/1	5/6	51	23	6	52	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:49 no_useful_candidate:34 duplicate:6 exercise_set_cap:5 near_equivalent:1	1/1	5/6	50	23	6	52	24	8	template_reference_5	5

min 75 — advanced — p31-5-day-upper-priority-iranian-advanced .39

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:98 no_useful_candidate:68 exercise_set_cap:11 duplicate:11	2/2	6/8	67	28	7	58	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:98 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:4	2/2	6/8	65	28	7	56	26	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	0/0	75	32	8	61	27	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:147 no_useful_candidate:102 weekly_hard_volume_cap:36 exercise_set_cap:18 duplicate:18 acceptable_volume_preference:18 near_equivalent:7	3/3	7/10	68	31	8	52	24	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	weekly_hard_ volume_cap: 418 acceptable_v olume_prefer ence:201 wrong_templa te_muscle: 147 no_useful_ca ndidate:102 duplicate:21 exercise_set _cap:12 near_equival ent:3	3/3	5/8	66	29	8	52	24	8	template_ref erence_5	5

min 90 — advanced — p31-5-day-upper-priority-iranian-advanced .40

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REduced_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:196 no_useful_candidate:136 exercise_set_cap:26 duplicate:26	4/4	6/10	82	35	9	61	27	8	template_reference_1	1
DAY_LIMITS_0 K	weekly_hard_volume_cap:436 acceptable_volume_preference:218 wrong_template_muscle:196 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:10	4/4	8/12	81	36	9	56	26	8	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle:151 no_useful_candidate:34 near_equivalent:11 duplicate:9 exercise_set_cap:8	1/1	0/1	81	35	9	75	32	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	weekly_hard_volume_cap:355 wrong_template_muscle:245 acceptable_volume_preference:177 no_useful_candidate:170 duplicate:35 exercise_set_cap:34 near_equivalent:12	5/5	8/13	83	38	10	52	24	8	template_reference_4	4
UNDER_TARGET	weekly_hard_volume_cap:1081 acceptable_volume_preference:536 wrong_template_muscle:196 no_useful_candidate:134 duplicate:24 exercise_set_cap:6 near_equivalent:4	0/4	3/5	39	18	5	52	24	8	template_reference_5	5

min 30 — intermediate — p34-5-day-fst7-arms-priority-intermediate .41

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
 weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
 duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
 validation:failed[RECOVERY_SPACING_INVALID]
 final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	33	14	4	28	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	33	14	4	28	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	37	16	4	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	33	14	4	28	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	1/1	20	10	3	24	12	4	template_ref erence_5	5

min 45 — intermediate — p34-5-day-fst7-arms-priority-intermediate .42

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	39	17	5	40	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	41	18	5	40	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	53	24	6	40	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	1/1	37	17	5	40	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	3/3	36	18	5	39	18	6	template_ref erence_5	5

min 60 — intermediate — p34-5-day-fst7-arms-priority-intermediate .43

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:46 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:1	1/1	3/4	51	23	6	52	24	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:46 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:2	1/1	3/4	51	23	6	52	24	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	0/0	57	26	8	52	24	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:144 no_useful_candidate:34 exercise_set_cap:5 duplicate:5	1/1	6/7	51	24	6	52	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:212 no_useful_candidate:68 weekly_hard_volume_cap:36 acceptable_volume_preference:18 exercise_set_cap:11 duplicate:11 near_equivalent:6	2/2	6/8	54	27	7	53	24	8	template_reference_5	5

min 75 — intermediate — p34-5-day-fst7-arms-priority-intermediate .44

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:138 no_useful_candidate:102 exercise_set_cap:18 duplicate:18 near_equivalent:4	3/3	3/6	70	31	8	60	28	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:138 no_useful_candidate:102 exercise_set_cap:18 duplicate:18 near_equivalent:6	3/3	5/8	71	31	8	56	26	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	3/3	65	30	8	58	27	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:576 no_useful_candidate:136 weekly_hard_volume_cap:34 duplicate:26 exercise_set_cap:18 acceptable_volume_preference:7	4/4	6/10	69	33	9	52	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:424 weekly_hard_volume_cap:159 no_useful_candidate:136 acceptable_volume_preference:79 duplicate:26 exercise_set_cap:25 near_equivalent:12	4/4	7/11	68	34	9	53	24	8	template_reference_5	5

min 90 — intermediate — p34-5-day-fst7-arms-priority-intermediate .45

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REduced_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0K	wrong_template_muscle:184 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:6	4/4	4/8	80	35	9	65	30	8	template_reference_1	1
DAY_LIMITS_0K	weekly_hard_volume_cap:321 wrong_template_muscle:230 no_useful_candidate:170 acceptable_volume_preference:158 duplicate:35 exercise_set_cap:32 near_equivalent:14	5/5	6/11	85	38	10	61	28	8	template_reference_2	2
DAY_LIMITS_0K	wrong_template_muscle:294 no_useful_candidate:68 near_equivalent:18 exercise_set_cap:17 duplicate:17	2/2	6/8	83	39	10	61	28	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:1152 weekly_hard_volume_cap:308 no_useful_candidate:272 acceptable_volume_preference:141 duplicate:62 exercise_set_cap:30	4/8	6/12	69	33	9	52	24	8	template_reference_4	4
UNDER_TARGET	weekly_hard_volume_cap:632 wrong_template_muscle:530 acceptable_volume_preference:312 no_useful_candidate:170 duplicate:29 near_equivalent:15 exercise_set_cap:14	1/5	5/8	46	23	6	53	24	8	template_reference_5	5

min 30 — advanced — p35-5-day-fst7-arms-priority-advanced .46

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	33	14	4	28	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	35	15	4	28	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	37	16	4	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	30	13	4	28	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	24	12	4	24	12	4	template_ref erence_5	5

min 45 — advanced — p35-5-day-fst7-arms-priority-advanced .47

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage
RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant
template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	47	20	6	41	18	6	template_ref erence_1	1

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	41	18	6	41	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	55	24	6	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	38	17	5	40	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	3/3	36	18	5	39	18	6	template_ref erence_5	5

min 60 — advanced — p35-5-day-fst7-arms-priority-advanced .48

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARG
ET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESS
ION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	54	23	7	55	24	8	template_ref erence_1	1

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	2/2	51	24	7	52	24	8	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	64	29	8	53	24	8	template_ref erence_3	3
DAY_LIMITS_0 K	wrong_templa te_muscle: 149 no_useful_ca ndidate:34 exercise_set _cap:5 duplicate:5	1/1	5/6	51	23	6	53	24	8	template_ref erence_4	4
UNDER_TARGET	wrong_templa te_muscle: 700 weekly_hard_ volume_cap: 356 acceptable_v olume_prefer ence:178 no_useful_ca ndidate:170 duplicate:29 exercise_set _cap:17 near_equival ent:15	1/5	6/9	48	24	6	53	24	8	template_ref erence_5	5

min 75 — advanced — p35-5-day-fst7-arms-priority-advanced .49

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARG

ET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	4/4	65	28	7	57	25	8	template_ref erence_1	1
DAY_LIMITS_0 K	wrong_templa te_muscle:49 no_useful_ca ndidate:34 exercise_set _cap:7 duplicate:7 near_equival ent:3	1/1	3/4	66	31	8	58	27	8	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	68	31	8	63	29	8	template_ref erence_3	3
DAY_LIMITS_0 K	wrong_templa te_muscle: 447 no_useful_ca ndidate:102 exercise_set _cap:18 duplicate:18 weekly_hard volume_cap:6 acceptable_v olume_prefer ence:3	3/3	6/9	67	31	8	55	25	8	template_ref erence_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:700 weekly_hard_volume_cap:356 acceptable_volume_preference:178 no_useful_candidate:170 duplicate:29 exercise_set_cap:17 near_equivalent:15	1/5	6/9	48	24	6	53	24	8	template_reference_5	5

min 90 — advanced — p35-5-day-fst7-arms-priority-advanced .50

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:98 no_useful_candidate:68 exercise_set_cap:15 duplicate:15 near_equivalent:6	2/2	4/6	81	35	9	65	28	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:147 no_useful_candidate:102 weekly_hard_volume_cap:87 acceptable_volume_preference:43 duplicate:24 exercise_set_cap:23 near_equivalent:9	3/3	5/8	82	38	10	58	27	8	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle:302 no_useful_candidate:68 near_equivalent:22 exercise_set_cap:17 duplicate:17	2/2	1/3	85	39	10	71	32	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:745 weekly_hard_volume_cap:274 no_useful_candidate:170 acceptable_volume_preference:137 duplicate:29 exercise_set_cap:17 near_equivalent:4	1/5	5/8	53	24	6	55	25	8	template_reference_4	4
UNDER_TARGET	wrong_template_muscle:700 weekly_hard_volume_cap:374 acceptable_volume_preference:178 no_useful_candidate:170 duplicate:29 near_equivalent:15 exercise_set_cap:8	1/5	3/6	42	21	6	53	24	8	template_reference_5	5

min 30 — intermediate — p36-5-day-professional-compound-intermediate .51

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
 weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
 duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
 validation:failed[RECOVERY_SPACING_INVALID]
 final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	37	16	4	28	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	39	16	4	29	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	30	13	4	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	29	13	4	27	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	38	15	4	30	12	4	template_ref erence_5	5

min 45 — intermediate — p36-5-day-professional-compound-intermediate .52

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARG
ET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	40	18	6	40	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	40	17	5	41	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	40	18	6	40	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	weekly_hard_ volume_cap:4	0/0	1/1	35	16	5	39	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	46	19	6	43	18	6	template_ref erence_5	5

min 60 — intermediate — p36-5-day-professional-compound-intermediate .53

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARG
ET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	weekly_hard_ volume_cap:2	0/0	2/2	50	23	7	52	24	8	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	1/1	50	22	6	53	24	8	template_ref erence_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	52	24	8	52	24	8	template_ref erence_3	3
DAY_LIMITS_0 K	wrong_templa te_muscle: 298 no_useful_ca ndidate:68 weekly_hard_ volume_cap: 24 duplicate:11 exercise_set _cap:3 acceptable_v olume_prefer ence:2	2/2	3/5	51	24	7	53	24	8	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	58	25	8	55	24	8	template_ref erence_5	5

min 75 — intermediate — p36-5-day-professional-compound-intermediate .54

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARG
ET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESS
ION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:46 no_useful_candidate:34 duplicate:7 exercise_set_cap:6 weekly_hard_volume_cap:4 near_equivalent:2	1/1	3/4	65	30	8	58	27	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:294 no_useful_candidate:68 near_equivalent:18 duplicate:15 exercise_set_cap:13	2/2	4/6	69	31	8	57	26	8	template_reference_2	2
DAY_LIMITS_0 K	weekly_hard_volume_cap:6	0/0	6/6	66	31	8	54	25	8	template_reference_3	3
UNDER_TARGET	wrong_template_muscle:894 weekly_hard_volume_cap:238 no_useful_candidate:204 acceptable_volume_preferance:106 duplicate:39 exercise_set_cap:7	2/6	2/6	49	23	7	53	24	8	template_reference_4	4
DAY_LIMITS_0 K	weekly_hard_volume_cap:8	0/0	4/4	67	29	8	55	24	8	template_reference_5	5

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_EFFECTIVE_TARGET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:138 no_useful_candidate:102 duplicate:24 exercise_set_cap:21 weekly_hard_volume_cap:12 near_equivalent:6 acceptable_volume_preference:2	3/3	5/8	84	38	10	58	27	8	template_reference_1	1

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:588 no_useful_candidate:136 near_equivalent:36 duplicate:34 exercise_set_cap:30	4/4	6/10	86	39	10	60	27	8	template_reference_2	2
DAY_LIMITS_0 K	weekly_hard_volume_cap:125 wrong_template_muscle:92 no_useful_candidate:68 acceptable_volume_preference:58 duplicate:17 exercise_set_cap:15 near_equivalent:10	2/2	7/9	84	38	10	54	25	8	template_reference_3	3
UNDER_TARGET	wrong_template_muscle:894 weekly_hard_volume_cap:238 no_useful_candidate:204 acceptable_volume_preference:106 duplicate:39 exercise_set_cap:7	2/6	2/6	49	23	7	53	24	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	weekly_hard_volume_cap:292 acceptable_volume_preference:138 no_useful_candidate:68 wrong_template_muscle:22 near_equivalent:20 duplicate:17 exercise_set_cap:13	2/2	6/8	86	37	10	57	25	8	template_reference_5	5

min 30 — advanced — p37-5-day-professional-compound-advanced .56

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	35	15	4	28	12	4	template_reference_1	1
DAY_LIMITS_0 K	—	0/0	0/0	39	16	4	29	12	4	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	30	13	4	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	30	13	4	28	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	39	15	4	30	12	4	template_ref erence_5	5

min 45 — advanced — p37-5-day-professional-compound-advanced .57

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	49	21	6	41	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	55	24	6	41	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	1/1	36	16	5	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	1/1	35	16	5	39	18	6	template_ref erence_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	51	21	6	42	18	6	template_ref erence_5	5

min 60 — advanced — p37-5-day-professional-compound-advanced .58

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	54	23	7	55	24	8	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	66	29	8	54	24	8	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	3/3	51	24	7	52	24	8	template_ref erence_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:770 weekly_hard_volume_cap:231 no_useful_candidate:170 acceptable_volume_preference:113 duplicate:29 exercise_set_cap:12 near_equivalent:4	1/5	4/7	47	22	6	53	24	8	template_reference_4	4
DAY_LIMITS_OK	—	0/0	0/0	60	26	8	54	24	8	template_reference_5	5

min 75 — advanced — p37-5-day-professional-compound-advanced .59

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	4/4	65	28	7	57	25	8	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	66	29	8	60	27	8	template_ref erence_2	2
DAY_LIMITS_0 K	wrong_templa te_muscle:49 no_useful_ca ndidate:34 exercise_set _cap:7 duplicate:7 near_equival ent:4	1/1	4/5	67	31	8	58	27	8	template_ref erence_3	3
UNDER_TARGET	wrong_templa te_muscle: 616 weekly_hard volume_cap: 248 no_useful_ca ndidate:136 acceptable_v olume_prefer ence:120 duplicate:20 exercise_set _cap:6	0/4	2/4	39	18	5	55	25	8	template_ref erence_4	4
DAY_LIMITS_0 K	weekly_hard_ volume_cap:3	0/0	3/3	66	29	8	54	24	8	template_ref erence_5	5

min 90 — advanced — p37-5-day-professional-compound-advanced .60

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
 duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]
 recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
 weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]
 duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]
 validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]
 final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:98 no_useful_candidate:68 exercise_set_cap:15 duplicate:15 near_equivalent:2	2/2	4/6	81	35	9	65	28	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:151 no_useful_candidate:34 near_equivalent:11 exercise_set_cap:8 duplicate:8	1/1	4/5	81	36	9	66	29	8	template_reference_2	2
DAY_LIMITS_0 K	weekly_hard_volume_cap:210 wrong_template_muscle:147 acceptable_volume_preferance:105 no_useful_candidate:102 exercise_set_cap:24 duplicate:24 near_equivalent:12	3/3	6/9	84	39	10	58	27	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:616 weekly_hard_volume_cap:248 no_useful_candidate:136 acceptable_volume_preference:120 duplicate:20 exercise_set_cap:6	0/4	2/4	39	18	5	55	25	8	template_reference_4	4
DAY_LIMITS_0K	weekly_hard_volume_cap:278 acceptable_volume_preference:132 no_useful_candidate:68 wrong_template_muscle:22 near_equivalent:18 duplicate:17 exercise_set_cap:13	2/2	5/7	83	37	10	59	26	8	template_reference_5	5

min 30 — intermediate — p38-6-day-ppl-ab-intermediate .61

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	29	12	4	29	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	30	13	4	28	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	29	13	4	27	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	22	9	3	28	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	24	10	3	29	12	4	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	36	15	4	29	12	4	template_ref erence_6	6

min 45 — intermediate — p38-6-day-ppl-ab-intermediate .62

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	41	18	6	41	18	6	template_ref erence_1	1

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	40	18	6	40	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	37	17	5	39	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	1/1	37	16	5	40	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	1/1	36	16	5	41	18	6	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	51	22	6	42	18	6	template_ref erence_6	6

min 60 — intermediate — p38-6-day-ppl-ab-intermediate .63

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:114 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:1	1/1	5/6	53	23	6	54	24	8	template_reference_1	1
DAY_LIMITS_0 K	—	0/0	5/5	51	23	6	53	24	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	6/6	51	24	6	51	24	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:114 no_useful_candidate:34 weekly_hard_volume_cap:5 duplicate:5 exercise_set_cap:4	1/1	4/5	51	22	6	53	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:87 no_useful_candidate:34 weekly_hard_volume_cap:6 duplicate:5 exercise_set_cap:4 near_equivalent:2	1/1	5/6	51	23	6	53	24	8	template_reference_5	5
DAY_LIMITS_0 K	—	0/0	3/3	51	22	6	54	24	8	template_reference_6	6

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:228 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:2	2/2	6/8	65	28	7	58	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:162 no_useful_candidate:68 exercise_set_cap:13 duplicate:13 near_equivalent:4	2/2	6/8	70	31	8	57	26	8	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:294 no_useful_candidate:68 exercise_set_cap:13 duplicate:13 near_equivalent:10	2/2	6/8	65	31	8	55	26	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:342 no_useful_candidate:102 weekly_hard_volume_cap:48 acceptable_volume_preference:24 exercise_set_cap:18 duplicate:18 near_equivalent:4	3/3	7/10	69	31	8	53	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:261 no_useful_candidate:102 weekly_hard_volume_cap:85 acceptable_volume_preference:42 duplicate:18 exercise_set_cap:17 near_equivalent:6	3/3	6/9	66	30	8	53	24	8	template_reference_5	5

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:147 no_useful_candidate:34 near_equivalent:9 weekly_hard_volume_cap:8 exercise_set_cap:6 duplicate:6 acceptable_volume_preference:4	1/1	6/7	67	28	7	54	24	8	template_reference_6	6

min 90 — intermediate — p38-6-day-ppl-ab-intermediate .65

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 456 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:8	4/4	7/11	82	35	9	58	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle: 243 no_useful_candidate:102 exercise_set_cap:21 duplicate:21 near_equivalent:6	3/3	7/10	81	35	9	57	26	8	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle: 588 no_useful_candidate:136 exercise_set_cap:30 duplicate:30 near_equivalent:20 weekly_hard_volume_cap:4 acceptable_volume_preference:2	4/4	8/12	84	39	10	55	26	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle: 570 weekly_hard_volume_cap: 252 no_useful_candidate:170 acceptable_volume_preference:118 duplicate:35 exercise_set_cap:28	5/5	8/13	81	38	10	53	24	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	weekly_hard_volume_cap: 865 wrong_template_muscle: 609 acceptable_volume_preference: 424 no_useful_candidate: 238 duplicate: 50 exercise_set_cap: 25 near_equivalent: 14	3/7	6/11	65	30	8	53	24	8	template_reference_5	5
DAY_LIMITS_OK	wrong_template_muscle: 441 no_useful_candidate: 102 weekly_hard_volume_cap: 42 near_equivalent: 27 exercise_set_cap: 21 acceptable_volume_preference: 21 duplicate: 21	3/3	8/11	81	36	9	57	25	8	template_reference_6	6

min 45 — advanced — p39-6-day-ppl-ab-advanced .66

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	1/1	36	16	5	40	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	35	15	5	42	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	44	19	6	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	35	15	5	41	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	1/1	35	16	5	40	18	6	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	41	18	6	41	18	6	template_ref erence_6	6

min 60 — advanced — p39-6-day-ppl-ab-advanced .67

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage
RECOVERY_SPACING_INVALID :Reason codes
recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant
template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:119 no_useful_candidate:34 exercise_set_cap:5 duplicate:5	1/1	5/6	52	23	6	54	24	8	template_reference_1	1
DAY_LIMITS_0 K	—	0/0	4/4	51	22	6	54	24	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	0/0	56	25	8	53	24	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:119 no_useful_candidate:34 exercise_set_cap:5 duplicate:5	1/1	5/6	54	23	6	54	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:90 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:1	1/1	5/6	50	23	6	52	24	8	template_reference_5	5
DAY_LIMITS_0 K	wrong_template_muscle:151 no_useful_candidate:34 near_equivalent:11 exercise_set_cap:5 duplicate:5	1/1	4/5	55	23	6	55	24	8	template_reference_6	6

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCE_SET,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCE]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:238 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:2	2/2	6/8	65	28	7	58	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:84 no_useful_candidate:34 exercise_set_cap:6 duplicate:6 near_equivalent:2	1/1	6/7	65	28	7	60	27	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	3/3	65	29	8	57	26	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:238 no_useful_candidate:68 weekly_hard_volume_cap:26 acceptable_volume_preference:13 exercise_set_cap:11 duplicate:11 near_equivalent:2	2/2	7/9	65	28	7	54	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:270 no_useful_candidate:102 weekly_hard_volume_cap:86 acceptable_volume_preference:43 exercise_set_cap:18 duplicate:18 near_equivalent:3	3/3	7/10	69	31	8	52	24	8	template_reference_5	5
DAY_LIMITS_0 K	wrong_template_muscle:302 no_useful_candidate:68 near_equivalent:22 exercise_set_cap:11 duplicate:11	2/2	6/8	65	28	7	55	24	8	template_reference_6	6

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:476 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:6	4/4	7/11	82	35	9	58	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:252 no_useful_candidate:102 exercise_set_cap:21 duplicate:21 near_equivalent:6	3/3	7/10	80	35	9	62	28	8	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:302 no_useful_candidate:68 near_equivalent:24 exercise_set_cap:17 duplicate:17	2/2	7/9	85	39	10	60	27	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:595 weekly_hard_volume_cap:250 no_useful_candidate:170 acceptable_volume_preference:118 duplicate:35 exercise_set_cap:28 near_equivalent:9	5/5	7/12	80	37	10	54	24	8	template_reference_4	4
UNDER_TARGET	weekly_hard_volume_cap:913 wrong_template_muscle:630 acceptable_volume_preference:448 no_useful_candidate:238 duplicate:50 exercise_set_cap:25 near_equivalent:7	3/7	6/11	64	30	8	52	24	8	template_reference_5	5

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:604 no_useful_candidate:136 near_equivalent:44 weekly_hard_volume_cap:34 exercise_set_cap:26 duplicate:26 acceptable_volume_preference:17	4/4	7/11	85	35	9	58	25	8	template_reference_6	6

min 90 — intermediate — p40-6-day-upper-lower-x3-intermediate .70

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

SEMANTIC_OPENER_CONFLICT

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REduced_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SEMANTIC_OPENER_CONFLICT]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SEMANTIC_OPENER_CONFLICT,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:138 no_useful_candidate:102 exercise_set_cap:21 duplicate:21 near_equivalent:10	3/3	6/9	80	35	9	60	27	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:735 no_useful_candidate:170 weekly_hard_volume_cap:52 exercise_set_cap:35 duplicate:35 acceptable_volume_preference:26 near_equivalent:2	5/5	9/14	83	40	10	55	26	8	template_reference_2	2
DAY_LIMITS_0 K	weekly_hard_volume_cap:282 wrong_template_muscle:184 acceptable_volume_preference:141 no_useful_candidate:136 exercise_set_cap:30 duplicate:30 near_equivalent:13	4/4	8/12	85	39	10	55	25	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:1176 weekly_hard_volume_cap:284 no_useful_candidate:272 acceptable_volume_preference:136 duplicate:62 near_equivalent:40 exercise_set_cap:36	4/8	6/12	75	34	9	56	25	8	template_reference_4	4
DAY_LIMITS_0K	weekly_hard_volume_cap:550 acceptable_volume_preference:269 wrong_template_muscle:184 no_useful_candidate:136 duplicate:30 exercise_set_cap:25 near_equivalent:8	4/4	7/11	84	37	10	54	24	8	template_reference_5	5
UNDER_TARGET	wrong_template_muscle:588 weekly_hard_volume_cap:174 no_useful_candidate:134 acceptable_volume_preference:84 near_equivalent:36 duplicate:24 exercise_set_cap:6	0/4	1/3	44	18	5	61	26	8	template_reference_6	6

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SAFE_TEMPLATE_SUPERSET_PRESERVED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:196 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:4	4/4	5/9	82	35	9	62	27	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:453 no_useful_candidate:102 near_equivalent:36 exercise_set_cap:21 duplicate:21 weekly_hard_volume_cap:4 acceptable_volume_preference:2	3/3	7/10	81	36	9	60	27	8	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	weekly_hard_volume_cap:444 acceptable_volume_preference:222 wrong_template_muscle:196 no_useful_candidate:136 exercise_set_cap:30 duplicate:30 near_equivalent:10	4/4	7/11	86	39	10	61	28	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:604 no_useful_candidate:136 weekly_hard_volume_cap:44 near_equivalent:44 exercise_set_cap:26 duplicate:26 acceptable_volume_preference:22	4/4	8/12	81	35	9	54	24	8	template_reference_4	4
DAY_LIMITS_0 K	weekly_hard_volume_cap:558 acceptable_volume_preference:274 wrong_template_muscle:196 no_useful_candidate:136 duplicate:30 exercise_set_cap:25 near_equivalent:9	4/4	6/10	84	37	10	56	25	8	template_reference_5	5

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:755 weekly_hard_volume_cap:194 no_useful_candidate:168 acceptable_volume_preference:94 near_equivalent:60 duplicate:34 exercise_set_cap:14	1/5	5/8	53	23	6	53	24	8	template_reference_6	6

min 30 — intermediate — p42-6-day-fitclub-hybrid-intermediate .72

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0_K	—	0/0	0/0	32	14	4	28	12	4	template_reference_1	1
DAY_LIMITS_0_K	—	0/0	0/0	23	10	3	28	12	4	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	29	13	4	27	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	31	14	4	27	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	30	12	4	30	12	4	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	36	15	4	29	12	4	template_ref erence_6	6

min 45 — intermediate — p42-6-day-fitclub-hybrid-intermediate .73

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	41	18	6	41	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	38	17	5	41	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	35	16	5	39	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	weekly_hard volume_cap:3	0/0	1/1	35	16	5	39	18	6	template_ref erence_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	42	18	6	42	18	6	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	48	21	6	41	18	6	template_ref erence_6	6

min 60 — intermediate — p42-6-day-fitclub-hybrid-intermediate .74

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	4/4	51	22	6	55	24	8	template_ref erence_1	1
DAY_LIMITS_0 K	wrong_templa te_muscle: 114 no_useful_ca ndidate:34 duplicate:5 weekly_hard_ volume_cap:4 exercise_set _cap:4 near_equival ent:3	1/1	3/4	52	23	6	45	20	6	template_ref erence_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	3/3	51	24	7	51	24	8	template_ref erence_3	3
DAY_LIMITS_0 K	wrong_templa te_muscle: 298 no_useful_ca ndidate:68 weekly_hard_ volume_cap: 23 duplicate:13 exercise_set _cap:5 acceptable_v olume_prefer ence:4	2/2	3/5	51	24	7	53	24	8	template_ref erence_4	4
DAY_LIMITS_0 K	weekly_hard_ volume_cap:6	0/0	3/3	50	21	6	56	24	8	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	3/3	50	22	6	53	24	8	template_ref erence_6	6

min 75 — intermediate — p42-6-day-fitclub-hybrid-intermediate .75

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:342 no_useful_candidate:102 exercise_set_cap:18 duplicate:18 near_equivalent:5	3/3	5/8	70	31	8	59	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:342 no_useful_candidate:102 duplicate:18 exercise_set_cap:15 near_equivalent:9 weekly_hard_volume_cap:7	3/3	4/7	70	30	8	45	20	6	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle:147 no_useful_candidate:34 exercise_set_cap:7 duplicate:7 near_equivalent:5	1/1	5/6	65	31	8	55	26	8	template_reference_3	3
UNDER_TARGET	wrong_template_muscle:894 weekly_hard_volume_cap:240 no_useful_candidate:202 acceptable_volume_preference:108 duplicate:45 exercise_set_cap:13	2/6	4/8	53	25	7	53	24	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	weekly_hard_volume_cap:726 wrong_template_muscle:475 acceptable_volume_preference:355 no_useful_candidate:170 duplicate:34 exercise_set_cap:12 near_equivalent:10	1/5	4/7	59	25	7	56	24	8	template_reference_5	5
DAY_LIMITS_0K	wrong_template_muscle:147 no_useful_candidate:34 duplicate:7 exercise_set_cap:6 weekly_hard_volume_cap:6 near_equivalent:5 acceptable_volume_preference:3	1/1	6/7	65	28	7	53	24	8	template_reference_6	6

min 90 — intermediate — p42-6-day-fitclub-hybrid-intermediate .76

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]
validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:456 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:8	4/4	6/10	80	35	9	59	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:456 no_useful_candidate:136 weekly_hard_volume_cap:86 acceptable_volume_preferance:38 duplicate:26 exercise_set_cap:22 near_equivalent:12	4/4	6/10	80	35	9	45	20	6	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 441 no_useful_candidate:102 exercise_set_cap:24 duplicate:24 near_equivalent:15 weekly_hard_volume_cap:8 acceptable_volume_preference:4	3/3	7/10	84	39	10	55	26	8	template_reference_3	3
UNDER_TARGET	wrong_template_muscle: 894 weekly_hard_volume_cap: 240 no_useful_candidate:202 acceptable_volume_preference:108 duplicate:45 exercise_set_cap:13	2/6	4/8	53	25	7	53	24	8	template_reference_4	4
UNDER_TARGET	weekly_hard_volume_cap: 645 wrong_template_muscle: 380 acceptable_volume_preference:316 no_useful_candidate:136 duplicate:24 near_equivalent:8 exercise_set_cap:4	0/4	2/4	47	20	6	56	24	8	template_reference_5	5

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:588 no_useful_candidate:136 weekly_hard_volume_cap:52 duplicate:34 exercise_set_cap:30 acceptable_volume_preference:26 near_equivalent:21	4/4	8/12	85	39	10	56	25	8	template_reference_6	6

min 30 — advanced — p43-6-day-fitclub-hybrid-advanced .77

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,TEMPLATE_SUPERSET_REJECTED_UNSAFE]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	30	13	4	28	12	4	template_reference_1	1
DAY_LIMITS_0 K	—	0/0	0/0	24	10	3	28	12	4	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	37	16	4	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	27	12	4	27	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	30	12	4	30	12	4	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	37	15	4	29	12	4	template_ref erence_6	6

min 45 — advanced — p43-6-day-fitclub-hybrid-advanced .78

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,TEMPLATE_SUPERSET_REJECTED_UNSAFE]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	41	18	6	41	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	38	17	5	41	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	48	21	6	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	1/1	35	16	5	39	18	6	template_ref erence_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	42	18	6	42	18	6	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	37	16	5	40	18	6	template_ref erence_6	6

min 60 — advanced — p43-6-day-fitclub-hybrid-advanced .79

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,TEMPLATE_SUPERSET_REJECTED_UNSAFE]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	4/4	51	22	6	55	24	8	template_ref erence_1	1
DAY_LIMITS_0 K	wrong_templa te_muscle: 119 no_useful_ca ndidate:34 exercise_set _cap:5 duplicate:5 near_equivale nt:3	1/1	3/4	52	23	6	45	20	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	54	24	7	53	24	8	template_ref erence_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:154 no_useful_candidate:34 duplicate:6 exercise_set_cap:5 weekly_hard_volume_cap:2 acceptable_volume_preference:1 near_equivalent:1	1/1	6/7	51	24	6	53	24	8	template_reference_4	4
DAY_LIMITS_0 K	—	0/0	3/3	50	21	6	56	24	8	template_reference_5	5
DAY_LIMITS_0 K	—	0/0	3/3	51	23	6	53	24	8	template_reference_6	6

min 75 — advanced — p43-6-day-fitclub-hybrid-advanced .80

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,TEMPLATE_SUPERSET_REJECTED_UNSAFE]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:238 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:2	2/2	5/7	65	28	7	59	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:238 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:6	2/2	4/6	65	28	7	47	21	6	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	1/1	67	30	8	59	27	8	template_reference_3	3
UNDER_TARGET	wrong_template_muscle:924 weekly_hard_volume_cap:224 no_useful_candidate:202 acceptable_volume_preference:112 duplicate:45 exercise_set_cap:25 near_equivalent:6	2/6	6/10	59	28	7	55	25	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	weekly_hard_volume_cap:106 wrong_template_muscle:98 acceptable_volume_preference:44 no_useful_candidate:34 duplicate:7 exercise_set_cap:3 near_equivalent:1	1/1	4/5	67	28	8	56	24	8	template_reference_5	5
DAY_LIMITS_0 K	wrong_template_muscle:151 no_useful_candidate:34 duplicate:8 exercise_set_cap:7 near_equivalent:1	1/1	5/6	67	31	8	53	24	8	template_reference_6	6

min 90 — advanced — p43-6-day-fitclub-hybrid-advanced .81

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED,TEMPLATE_SUPerset_REJECTED_UNSAFE]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:476 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:6	4/4	6/10	82	35	9	59	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:476 weekly_hard_volume_cap:168 no_useful_candidate:136 acceptable_volume_preference:84 exercise_set_cap:26 duplicate:26 near_equivalent:12	4/4	6/10	81	36	9	47	21	6	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle:302 no_useful_candidate:68 near_equivalent:24 exercise_set_cap:17 duplicate:17 weekly_hard_volume_cap:4 acceptable_volume_preference:2	2/2	4/6	86	39	10	64	29	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:924 weekly_hard_volume_cap:224 no_useful_candidate:202 acceptable_volume_preference:112 duplicate:45 exercise_set_cap:25 near_equivalent:6	2/6	6/10	59	28	7	55	25	8	template_reference_4	4
UNDER_TARGET	weekly_hard_volume_cap:680 wrong_template_muscle:392 acceptable_volume_preference:332 no_useful_candidate:136 duplicate:28 exercise_set_cap:4 near_equivalent:4	0/4	2/4	54	23	7	56	24	8	template_reference_5	5
DAY_LIMITS_0K	wrong_template_muscle:302 no_useful_candidate:68 duplicate:17 exercise_set_cap:15 weekly_hard_volume_cap:12 acceptable_volume_preference:6 near_equivalent:2	2/2	7/9	81	36	9	59	26	8	template_reference_6	6

min 30 — intermediate — p44-6-day-arnold-split-intermediate .82

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	30	12	4	30	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	23	10	3	27	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	35	16	4	27	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	30	12	4	30	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	23	10	3	27	12	4	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	39	16	4	29	12	4	template_ref erence_6	6

min 45 — intermediate — p44-6-day-arnold-split-intermediate .83

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	42	18	6	42	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	35	16	5	39	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	37	17	5	39	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	36	15	5	43	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	35	16	5	39	18	6	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	51	22	6	42	18	6	template_ref erence_6	6

min 60 — intermediate — p44-6-day-arnold-split-intermediate .84

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage
RECOVERY_SPACING_INVALID :Reason codes
recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant
template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:95 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:2	1/1	5/6	55	23	6	55	24	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:106 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:2	1/1	6/7	51	24	6	51	24	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	6/6	51	24	6	51	24	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:116 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:2	1/1	4/5	55	23	6	52	22	7	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:106 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:1	1/1	6/7	51	24	6	51	24	8	template_reference_5	5
DAY_LIMITS_0 K	—	0/0	3/3	51	22	6	54	24	8	template_reference_6	6

min 75 — intermediate — p44-6-day-arnold-split-intermediate .85

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

SEMANTIC_OPENER_CONFLICT :Reason codes

RECOVERY_SPACING_INVALID

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[SEMANTIC_OPENER_CONFLICT,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SEMANTIC_OPENER_CONFLICT,RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:95 no_useful_candidate:34 exercise_set_cap:6 duplicate:6 near_equivalent:2	1/1	7/8	65	28	7	55	24	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:318 no_useful_candidate:102 weekly_hard_volume_cap:68 acceptable_volume_preference:34 exercise_set_cap:18 duplicate:18 near_equivalent:11	3/3	5/8	65	31	8	59	28	8	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:294 no_useful_candidate:68 exercise_set_cap:13 duplicate:13 near_equivalent:10	2/2	6/8	65	31	8	55	26	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:232 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:4 weekly_hard_volume_cap:4 acceptable_volume_preference:2	2/2	6/8	67	28	7	52	22	7	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:212 weekly_hard_volume_cap:186 acceptable_volume_preference:90 no_useful_candidate:68 duplicate:13 exercise_set_cap:11 near_equivalent:2	2/2	7/9	65	31	8	51	24	8	template_reference_5	5

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:147 no_useful_candidate:34 near_equivalent:9 weekly_hard_volume_cap:8 exercise_set_cap:6 duplicate:6 acceptable_volume_preference:4	1/1	6/7	67	28	7	54	24	8	template_reference_6	6

min 90 — intermediate — p44-6-day-arnold-split-intermediate .86

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 285 no_useful_candidate:102 exercise_set_cap:21 duplicate:21 near_equivalent:6	3/3	8/11	82	35	9	55	24	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle: 530 weekly_hard_volume_cap: 224 no_useful_candidate:170 acceptable_volume_preference:112 exercise_set_cap:35 duplicate:35 near_equivalent:20	5/5	7/12	81	39	10	59	28	8	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle: 588 no_useful_candidate:136 exercise_set_cap:30 duplicate:30 near_equivalent:20 weekly_hard_volume_cap:4 acceptable_volume_preference:2	4/4	8/12	84	39	10	55	26	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:464 weekly_hard_volume_cap:257 no_useful_candidate:136 acceptable_volume_preference:126 duplicate:26 exercise_set_cap:23 near_equivalent:8	4/4	6/10	80	34	9	52	22	7	template_reference_4	4
UNDER_TARGET	weekly_hard_volume_cap:564 wrong_template_muscle:424 acceptable_volume_preference:276 no_useful_candidate:136 duplicate:24 exercise_set_cap:8 near_equivalent:4	0/4	4/6	47	22	6	51	24	8	template_reference_5	5
DAY_LIMITS_0 K	wrong_template_muscle:441 no_useful_candidate:102 weekly_hard_volume_cap:42 near_equivalent:27 exercise_set_cap:21 acceptable_volume_preference:21 duplicate:21	3/3	8/11	81	36	9	57	25	8	template_reference_6	6

min 30 — advanced — p45-6-day-arnold-split-advanced .87

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SAFE_TEMPLATE_SUPERSET_PRESERVED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,TEMPLATE_SUPERSET_REJECTED_UNSAFE]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	26	12	4	30	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	23	10	3	27	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	37	16	4	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	30	12	4	30	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	27	14	4	27	12	4	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	39	16	4	29	12	4	template_ref erence_6	6

min 45 — advanced — p45-6-day-arnold-split-advanced .88

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SAFE_TEMPLATE_SUPERSET_PRESERVED,TEMPLATE_SUPERSET_REJECTED_UNSAFE,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	1/1	35	16	5	43	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	2/2	36	18	5	39	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	44	19	6	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	36	15	5	43	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	35	16	5	39	18	6	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	42	18	6	42	18	6	template_ref erence_6	6

min 60 — advanced — p45-6-day-arnold-split-advanced .89

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage
RECOVERY_SPACING_INVALID :Reason codes
recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant
template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SAFE_TEMPLATE_SUPERSET_PRESERVED,TEMPLATE_SUPERSET_REJECTED_UNSAFE,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:98 no_useful_candidate:34 exercise_set_cap:5 duplicate:5	1/1	5/6	52	23	6	50	21	7	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:222 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:8	2/2	6/8	54	27	7	51	24	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	0/0	51	22	7	54	24	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:98 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:1	1/1	5/6	56	23	6	50	21	7	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:111 weekly_hard_volume_cap:44 no_useful_candidate:34 acceptable_volume_preference:22 exercise_set_cap:5 duplicate:5 near_equivalent:2	1/1	6/7	51	24	6	52	24	8	template_reference_5	5
DAY_LIMITS_0 K	wrong_template_muscle:151 no_useful_candidate:34 near_equivalent:11 exercise_set_cap:5 duplicate:5	1/1	5/6	51	23	6	53	24	8	template_reference_6	6

min 75 — advanced — p45-6-day-arnold-split-advanced .90

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SAFE_TEMPLATE_SUPERSET_PRESERVED,TEMPLATE_SUPERSET_REJECTED_UNSAFE,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:196 no_useful_candidate:68 exercise_set_cap:11 duplicate:11	2/2	7/9	65	28	7	53	22	7	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:444 weekly_hard_volume_cap:150 no_useful_candidate:136 acceptable_volume_preference:75 exercise_set_cap:26 duplicate:26 near_equivalent:16	4/4	6/10	69	35	9	59	28	8	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	5/5	65	28	7	58	26	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:196 weekly_hard_volume_cap:88 no_useful_candidate:68 acceptable_volume_preference:44 exercise_set_cap:11 duplicate:11 near_equivalent:2	2/2	6/8	66	27	7	50	21	7	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	weekly_hard_volume_cap:654 wrong_template_muscle:555 acceptable_volume_preference:319 no_useful_candidate:170 duplicate:34 near_equivalent:15 exercise_set_cap:14	1/5	5/8	55	26	7	52	24	8	template_reference_5	5
DAY_LIMITS_0K	wrong_template_muscle:453 no_useful_candidate:102 near_equivalent:33 exercise_set_cap:18 duplicate:18 weekly_hard_volume_cap:4 acceptable_volume_preference:2	3/3	7/10	68	31	8	53	24	8	template_reference_6	6

min 90 — advanced — p45-6-day-arnold-split-advanced .91

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
 duration_repair:completed[SAFE_TEMPLATE_SUPERSET_PRESERVED,TEMPLATE_SUPERSET_REJECTED_UNSAFE,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]
 recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
 weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]
 duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]
 validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]
 final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:392 no_useful_candidate:136 exercise_set_cap:26 duplicate:26	4/4	8/12	80	35	9	53	22	7	template_reference_1	1
UNDER_TARGET	wrong_template_muscle:999 weekly_hard_volume_cap:736 acceptable_volume_preference:368 no_useful_candidate:306 duplicate:75 exercise_set_cap:55 near_equivalent:41	5/9	8/15	79	40	10	59	28	8	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle:302 no_useful_candidate:68 near_equivalent:24 exercise_set_cap:15 duplicate:15	2/2	7/9	81	36	9	61	27	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	weekly_hard_volume_cap: 929 wrong_template_muscle: 686 acceptable_volume_preference: 459 no_useful_candidate: 238 duplicate: 50 exercise_set_cap: 29 near_equivalent: 11	3/7	7/12	76	31	8	50	21	7	template_reference_4	4
UNDER_TARGET	weekly_hard_volume_cap: 564 wrong_template_muscle: 444 acceptable_volume_preference: 276 no_useful_candidate: 136 duplicate: 24 near_equivalent: 12 exercise_set_cap: 8	0/4	4/6	47	22	6	52	24	8	template_reference_5	5
DAY_LIMITS_0 K	wrong_template_muscle: 604 no_useful_candidate: 136 near_equivalent: 44 weekly_hard_volume_cap: 32 exercise_set_cap: 26 duplicate: 26 acceptable_volume_preference: 16	4/4	8/12	82	35	9	53	24	8	template_reference_6	6

min 30 — intermediate — p46-6-day-classic-body-part-intermediate .92

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	27	11	3	28	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	22	11	3	24	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	37	16	4	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	27	12	4	27	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	32	14	4	28	12	4	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	21	9	3	27	12	4	template_ref erence_6	6

min 45 — intermediate — p46-6-day-classic-body-part-intermediate .93

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :**Stage chain**
 session_building:completed
 volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
 duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]
 recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
 weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
 duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
 validation:failed[RECOVERY_SPACING_INVALID]
 final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	41	18	5	36	16	5	template_ref erence_1	1
DAY_LIMITS_0 K	wrong_templa te_muscle: 132 no_useful_ca ndidate:34 duplicate:4 near_equival ent:4 exercise_set _cap:3 weekly_hard_ volume_cap:2	1/1	1/2	36	18	5	34	17	5	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	53	24	6	40	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	3/3	36	18	5	39	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	41	18	5	41	18	6	template_ref erence_5	5

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:149 no_useful_candidate:34 weekly_hard_volume_cap:12 duplicate:4 acceptable_volume_preference:3 exercise_set_cap:2	1/1	2/3	37	17	5	39	18	6	template_reference_6	6

min 60 — intermediate — p46-6-day-classic-body-part-intermediate .94

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_EXERCISE_COUNT_OUT_OF_RANGE :Reason codes

SESSION_DURATION_UNDER_TARGET

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_EXERCISE_COUNT_OUT_OF_RANGE,SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_EXERCISE_COUNT_OUT_OF_RANGE,SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:140 no_useful_candidate:34 duplicate:5 exercise_set_cap:4 weekly_hard_volume_cap:2 near_equivalent:1	1/1	1/2	51	22	6	41	18	5	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:396 no_useful_candidate:102 weekly_hard_volume_cap:34 duplicate:15 near_equivalent:14 acceptable_volume_preference:14 exercise_set_cap:12	3/3	3/6	52	26	7	34	17	5	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	0/0	61	28	8	52	24	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:258 no_useful_candidate:68 weekly_hard_volume_cap:52 acceptable_volume_preference:26 exercise_set_cap:11 duplicate:11 near_equivalent:2	2/2	6/8	54	27	7	53	24	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:104 no_useful_candidate:34 weekly_hard_volume_cap:6 duplicate:5 exercise_set_cap:3 acceptable_volume_preference:1	1/1	1/2	51	22	6	41	18	6	template_reference_5	5
UNDER_TARGET COUNT_OUT_OF_RANGE	wrong_template_muscle:596 weekly_hard_volume_cap:240 no_useful_candidate:136 acceptable_volume_preference:116 duplicate:16	0/4	0/2	27	12	4	53	24	8	template_reference_6	6

min 75 — intermediate — p46-6-day-classic-body-part-intermediate .95

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

SESSION_EXERCISE_COUNT_OUT_OF_RANGE

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
 weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]
 duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]
 validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_EXERCISE_COUNT_OUT_OF_RANGE,RECOVERY_SPACING_INVALID]
 final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_EXERCISE_COUNT_OUT_OF_RANGE,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONS
 TRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:420 no_useful_candidate:102 duplicate:18 exercise_set_cap:15 near_equivalent:7 weekly_hard_volume_cap:6	3/3	3/6	70	30	8	41	18	5	template_reference_1	1
UNDER_TARGET	wrong_template_muscle:1056 weekly_hard_volume_cap:365 no_useful_candidate:272 acceptable_volume_preference:177 duplicate:54 near_equivalent:39 exercise_set_cap:32	4/8	5/11	62	31	8	34	17	5	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	2/2	65	30	8	58	27	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:645 weekly_hard_volume_cap:426 acceptable_volume_preference:210 no_useful_candidate:170 duplicate:29 exercise_set_cap:14 near_equivalent:5	1/5	3/6	46	23	6	57	26	8	template_reference_4	4
DAY_LIMITS_0_K	wrong_template_muscle:312 no_useful_candidate:102 weekly_hard_volume_cap:18 duplicate:18 exercise_set_cap:12 acceptable_volume_preference:4	3/3	2/5	69	29	8	41	18	6	template_reference_5	5
UNDER_TARGET COUNT_OUT_OF_RANGE	wrong_template_muscle:596 weekly_hard_volume_cap:240 no_useful_candidate:136 acceptable_volume_preference:116 duplicate:16	0/4	0/2	27	12	4	53	24	8	template_reference_6	6

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

SESSION_EXERCISE_COUNT_OUT_OF_RANGE

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_EXERCISE_COUNT_OUT_OF_RANGE,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,SESSION_EXERCISE_COUNT_OUT_OF_RANGE,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:560 no_useful_candidate:136 duplicate:26 exercise_set_cap:22 near_equivalent:10 weekly_hard_volume_cap:8	4/4	4/8	80	34	9	41	18	5	template_reference_1	1

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:1056 weekly_hard_volume_cap:365 no_useful_candidate:272 acceptable_volume_preference:177 duplicate:54 near_equivalent:39 exercise_set_cap:32	4/8	5/11	62	31	8	34	17	5	template_reference_2	2
DAY_LIMITS_0K	wrong_template_muscle:294 no_useful_candidate:68 near_equivalent:18 exercise_set_cap:17 duplicate:17	2/2	5/7	83	39	10	61	28	8	template_reference_3	3
UNDER_TARGET	wrong_template_muscle:645 weekly_hard_volume_cap:426 acceptable_volume_preference:210 no_useful_candidate:170 duplicate:29 exercise_set_cap:14 near_equivalent:5	1/5	3/6	46	23	6	57	26	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_OK	wrong_template_muscle:416 no_useful_candidate:136 weekly_hard_volume_cap:28 duplicate:26 exercise_set_cap:18 acceptable_volume_preference:6	4/4	4/8	81	34	9	41	18	6	template_reference_5	5
UNDER_TARGET_COUNT_OUT_OF_RANGE	wrong_template_muscle:596 weekly_hard_volume_cap:240 no_useful_candidate:136 acceptable_volume_preference:116 duplicate:16	0/4	0/2	27	12	4	53	24	8	template_reference_6	6

min 30 — advanced — p47-6-day-classic-body-part-advanced .97

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	27	11	3	28	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	22	11	3	24	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	39	16	4	29	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	27	12	4	27	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	29	12	4	29	12	4	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	23	10	3	27	12	4	template_ref erence_6	6

min 45 — advanced — p47-6-day-classic-body-part-advanced .98

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	43	18	5	35	15	5	template_ref erence_1	1

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	1/1	36	18	5	34	17	5	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	55	24	6	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	3/3	36	18	5	39	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	36	15	5	36	15	5	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	39	18	6	39	18	6	template_ref erence_6	6

min 60 — advanced — p47-6-day-classic-body-part-advanced .99

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:145 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:1	1/1	2/3	54	23	6	43	18	5	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:274 no_useful_candidate:68 duplicate:11 near_equivalent:10 exercise_set_cap:9 weekly_hard_volume_cap:5	2/2	3/5	52	26	7	34	17	5	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	0/0	69	29	8	55	24	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:134 no_useful_candidate:34 weekly_hard_volume_cap:8 duplicate:6 exercise_set_cap:5 acceptable_volume_preference:1 near_equivalent:1	1/1	5/6	55	26	7	53	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:142 no_useful_candidate:34 exercise_set_cap:5 duplicate:5	1/1	4/5	56	23	6	38	16	5	template_reference_5	5

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:616 weekly_hard_volume_cap:244 no_useful_candidate:136 acceptable_volume_preference:116 duplicate:24	0/4	0/2	40	18	6	53	24	8	template_reference_6	6

min 75 — advanced — p47-6-day-classic-body-part-advanced .100

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 290 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:3	2/2	4/6	67	28	7	43	18	5	template_reference_1	1
UNDER_TARGET	wrong_template_muscle: 822 weekly_hard_volume_cap: 320 no_useful_candidate:204 acceptable_volume_preference:160 duplicate:39 near_equivalent:30 exercise_set_cap:25	2/6	5/9	56	28	7	34	17	5	template_reference_2	2
DAY_LIMITS_0 K	—	0/0	0/0	75	32	8	61	27	8	template_reference_3	3
UNDER_TARGET	wrong_template_muscle: 670 weekly_hard_volume_cap: 440 acceptable_volume_preference:220 no_useful_candidate:170 duplicate:29 exercise_set_cap:17 near_equivalent:5	1/5	4/7	48	24	6	57	26	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_OK	wrong_template_muscle:284 no_useful_candidate:68 exercise_set_cap:11 duplicate:11	2/2	5/7	66	27	7	38	16	5	template_reference_5	5
UNDER_TARGET	wrong_template_muscle:616 weekly_hard_volume_cap:246 no_useful_candidate:136 acceptable_volume_preference:120 duplicate:20 exercise_set_cap:6	0/4	2/4	39	18	5	55	25	8	template_reference_6	6

min 90 — advanced — p47-6-day-classic-body-part-advanced .101

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:580 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:7	4/4	5/9	84	35	9	43	18	5	template_reference_1	1
UNDER_TARGET	wrong_template_muscle:822 weekly_hard_volume_cap:320 no_useful_candidate:204 acceptable_volume_preference:160 duplicate:39 near_equivalent:30 exercise_set_cap:25	2/6	5/9	56	28	7	34	17	5	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle:151 no_useful_candidate:34 near_equivalent:11 exercise_set_cap:8 duplicate:8	1/1	0/1	81	35	9	75	32	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
UNDER_TARGET	wrong_template_muscle:670 weekly_hard_volume_cap:440 acceptable_volume_preference:220 no_useful_candidate:170 duplicate:29 exercise_set_cap:17 near_equivalent:5	1/5	4/7	48	24	6	57	26	8	template_reference_4	4
UNDER_TARGET	wrong_template_muscle:852 no_useful_candidate:204 session_volume_cap:160 duplicate:39 exercise_set_cap:25	2/6	6/10	69	28	7	38	16	5	template_reference_5	5
UNDER_TARGET	wrong_template_muscle:616 weekly_hard_volume_cap:246 no_useful_candidate:136 acceptable_volume_preference:120 duplicate:20 exercise_set_cap:6	0/4	2/4	39	18	5	55	25	8	template_reference_6	6

min 30 — intermediate — p48-6-day-ronnie-double-exposure-intermediate .102

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	31	14	4	27	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	29	13	4	27	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	28	12	4	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	24	10	3	29	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	36	15	4	29	12	4	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	22	9	3	28	12	4	template_ref erence_6	6

min 45 — intermediate — p48-6-day-ronnie-double-exposure-intermediate .103

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	39	18	6	39	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	37	17	5	39	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	1/1	36	16	5	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	35	15	5	42	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	51	22	6	42	18	6	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	1/1	37	16	5	41	18	6	template_ref erence_6	6

min 60 — intermediate — p48-6-day-ronnie-double-exposure-intermediate .104

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage
RECOVERY_SPACING_INVALID :Reason codes
recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant
template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:87 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:1	1/1	5/6	50	23	6	52	24	8	template_reference_1	1
DAY_LIMITS_0 K	—	0/0	6/6	51	24	6	51	24	8	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle:114 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:1	1/1	5/6	51	23	6	55	24	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:87 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:3	1/1	5/6	53	23	6	54	24	8	template_reference_4	4
DAY_LIMITS_0 K	—	0/0	3/3	51	22	6	54	24	8	template_reference_5	5
DAY_LIMITS_0 K	wrong_template_muscle:114 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:2	1/1	5/6	51	23	6	55	24	8	template_reference_6	6

min 75 — intermediate — p48-6-day-ronnie-double-exposure-intermediate .105

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCED_SET,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:261 no_useful_candidate:102 exercise_set_cap:18 duplicate:18 near_equivalent:5	3/3	6/9	70	31	8	56	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle:294 no_useful_candidate:68 exercise_set_cap:13 duplicate:13 near_equivalent:10	2/2	6/8	65	31	8	55	26	8	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:342 no_useful_candidate:102 exercise_set_cap:18 duplicate:18 near_equivalent:5	3/3	6/9	70	31	8	59	26	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:174 weekly_hard_volume_cap:83 no_useful_candidate:68 acceptable_volume_preference:41 duplicate:13 exercise_set_cap:12 near_equivalent:6	2/2	6/8	67	30	8	54	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:147 no_useful_candidate:34 near_equivalent:9 weekly_hard_volume_cap:8 exercise_set_cap:6 duplicate:6 acceptable_volume_preference:4	1/1	6/7	67	28	7	54	24	8	template_reference_5	5

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:342 no_useful_candidate:102 weekly_hard_volume_cap:93 acceptable_volume_preference:45 duplicate:18 exercise_set_cap:16 near_equivalent:6	3/3	6/9	65	30	8	55	24	8	template_reference_6	6

min 90 — intermediate — p48-6-day-ronnie-double-exposure-intermediate .106

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REduced_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 348 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:7	4/4	7/11	81	35	9	56	26	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle: 588 no_useful_candidate:136 exercise_set_cap:30 duplicate:30 near_equivalent:20 weekly_hard_volume_cap:4 acceptable_volume_preference:2	4/4	8/12	84	39	10	55	26	8	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle: 456 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:8	4/4	7/11	80	35	9	59	26	8	template_reference_3	3

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	weekly_hard_volume_cap:415 wrong_template_muscle:348 acceptable_volume_preference:202 no_useful_candidate:136 duplicate:30 exercise_set_cap:26 near_equivalent:12	4/4	8/12	82	38	10	54	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:441 no_useful_candidate:102 weekly_hard_volume_cap:42 near_equivalent:27 exercise_set_cap:21 acceptable_volume_preference:21 duplicate:21	3/3	8/11	81	36	9	57	25	8	template_reference_5	5
UNDER_TARGET	wrong_template_muscle:570 weekly_hard_volume_cap:567 acceptable_volume_preference:277 no_useful_candidate:170 duplicate:29 exercise_set_cap:11 near_equivalent:10	1/5	4/7	49	22	6	55	24	8	template_reference_6	6

min 30 — advanced — p49-6-day-ronnie-double-exposure-advanced .107

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	34	14	4	29	12	4	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	37	16	4	28	12	4	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	0/0	32	14	4	28	12	4	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	24	10	3	29	12	4	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	39	16	4	29	12	4	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	0/0	32	14	4	28	12	4	template_ref erence_6	6

min 45 — advanced — p49-6-day-ronnie-double-exposure-advanced .108

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]
validation:failed[RECOVERY_SPACING_INVALID]
final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	—	0/0	0/0	37	16	5	41	18	6	template_ref erence_1	1
DAY_LIMITS_0 K	—	0/0	0/0	44	19	6	41	18	6	template_ref erence_2	2
DAY_LIMITS_0 K	—	0/0	1/1	36	16	5	41	18	6	template_ref erence_3	3
DAY_LIMITS_0 K	—	0/0	0/0	37	16	5	42	18	6	template_ref erence_4	4
DAY_LIMITS_0 K	—	0/0	0/0	41	18	6	41	18	6	template_ref erence_5	5
DAY_LIMITS_0 K	—	0/0	1/1	36	16	5	41	18	6	template_ref erence_6	6

min 60 — advanced — p49-6-day-ronnie-double-exposure-advanced .109

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage
RECOVERY_SPACING_INVALID :Reason codes
recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant
template_construction:succeeded :Stage chain
session_building:completed
volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCED]
duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]
recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]
weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]
duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:90 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:3	1/1	5/6	54	23	6	54	24	8	template_reference_1	1
DAY_LIMITS_0 K	—	0/0	0/0	56	25	8	53	24	8	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle:119 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:4	1/1	5/6	52	23	6	55	24	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:90 no_useful_candidate:34 exercise_set_cap:5 duplicate:5 near_equivalent:2	1/1	5/6	52	23	6	53	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:151 no_useful_candidate:34 near_equivalent:11 exercise_set_cap:5 duplicate:5	1/1	4/5	55	23	6	55	24	8	template_reference_5	5

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:119 no_useful_candidate:34 exercise_set_cap:5 duplicate:5	1/1	5/6	52	23	6	55	24	8	template_reference_6	6

min 75 — advanced — p49-6-day-ronnie-double-exposure-advanced .110

RECOVERY_SPACING_INVALID :Root cause | final_quality_gate :Failure stage

RECOVERY_SPACING_INVALID :Reason codes

recovery_spacing_is_valid() / repair_recovery_weekdays() :: app/workouts/program_engine/recovery.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REDUCE_SET,VOLUME_REPAIR_ADDED_SET_FOR_DIRECT_MINIMUM,VOLUME_REPAIR_SOFT_TARGET_REDUCE]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_ALREADY_BALANCED]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED]

validation:failed[RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[RECOVERY_SPACING_INVALID]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:180 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:4	2/2	6/8	65	28	7	58	26	8	template_reference_1	1
DAY_LIMITS_0 K	—	0/0	3/3	65	29	8	57	26	8	template_reference_2	2

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:238 no_useful_candidate:68 exercise_set_cap:11 duplicate:11 near_equivalent:8	2/2	6/8	65	28	7	59	26	8	template_reference_3	3
DAY_LIMITS_0 K	wrong_template_muscle:180 weekly_hard_volume_cap:88 no_useful_candidate:68 acceptable_volume_preference:44 exercise_set_cap:13 duplicate:13 near_equivalent:4	2/2	7/9	69	31	8	53	24	8	template_reference_4	4
DAY_LIMITS_0 K	wrong_template_muscle:302 no_useful_candidate:68 near_equivalent:22 exercise_set_cap:11 duplicate:11	2/2	6/8	65	28	7	55	24	8	template_reference_5	5

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:357 no_useful_candidate:102 weekly_hard_volume_cap:102 acceptable_volume_preference:51 exercise_set_cap:18 duplicate:18	3/3	7/10	69	31	8	55	24	8	template_reference_6	6

min 90 — advanced — p49-6-day-ronnie-double-exposure-advanced .111

SESSION_DURATION_UNDER_TARGET :Root cause | final_quality_gate :Failure stage

SESSION_DURATION_UNDER_TARGET :Reason codes

SESSION_DURATION_TARGET_UNSATISFIED

RECOVERY_SPACING_INVALID

SESSION_DURATION_CONSTRAINT_UNEXPLAINED

repair_session_durations() / _repair_underfill() :: app/workouts/program_engine/session_duration.py :Relevant

template_construction:succeeded :Stage chain

session_building:completed

volume_repair:completed[VOLUME_REPAIR_REMOVED_REDUNDANT_EXERCISE,VOLUME_REPAIR_REduced_SET,VOLUME_REPAIR_SOFT_TARGET_REDUCED]

duration_repair:completed[SESSION_DURATION_UNDERFILLED,SESSION_DURATION_REPAIR_APPLIED,SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

recovery_repair:completed[RECOVERY_WEEKDAY_REPAIR_UNAVAILABLE]

weekly_redistribution:completed[WEEKLY_REDISTRIBUTION_NO_SAFE_IMPROVING_MOVE]

duration_repair:completed[SESSION_DURATION_TARGET_SATISFIED,SESSION_DURATION_UNDERFILLED,SESSION_DURATION_CONSTRAINED_BY_HARD_VOLUME_LIMITS,SESSION_DURATION_TARGET_UNSATISFIED]

validation:failed[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID]

final_quality_gate:rejected[SESSION_DURATION_UNDER_TARGET,SESSION_DURATION_TARGET_UNSATISFIED,RECOVERY_SPACING_INVALID,SESSION_DURATION_CONSTRAINT_UNEXPLAINED]

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle: 360 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:8	4/4	6/10	82	35	9	60	27	8	template_reference_1	1
DAY_LIMITS_0 K	wrong_template_muscle: 302 no_useful_candidate:68 near_equivalent:24 exercise_set_cap:17 duplicate:17	2/2	7/9	85	39	10	60	27	8	template_reference_2	2
DAY_LIMITS_0 K	wrong_template_muscle: 476 no_useful_candidate:136 exercise_set_cap:26 duplicate:26 near_equivalent:16	4/4	7/11	82	35	9	59	26	8	template_reference_3	3
DAY_LIMITS_0 K	weekly_hard_volume_cap: 443 wrong_template_muscle: 360 acceptable_volume_preferance:216 no_useful_candidate:136 duplicate:30 exercise_set_cap:26 near_equivalent:8	4/4	8/12	81	38	10	53	24	8	template_reference_4	4

Day result	Candidate rejection categories	Exercise+	Set+	Final min	Final sets	Final count	Initial min	Initial sets	Initial count	Focus	Day
DAY_LIMITS_0 K	wrong_template_muscle:604 no_useful_candidate:136 near_equivalent:44 weekly_hard_volume_cap:34 exercise_set_cap:26 duplicate:26 acceptable_volume_preference:17	4/4	7/11	85	35	9	58	25	8	template_reference_5	5
UNDER_TARGET	weekly_hard_volume_cap:599 wrong_template_muscle:595 acceptable_volume_preference:293 no_useful_candidate:170 duplicate:29 exercise_set_cap:11 near_equivalent:4	1/5	4/7	48	22	6	55	24	8	template_reference_6	6

Competition Trace — Upper/Lower outputs .5

p32-5-day-upper-lower-specialty-intermediate → competition:5:intermediate:45

Professional candidates before first Upper/Lower rank: 4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p26-5-day-classic-body-part-intermediate	1
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p34-5-day-fst7-arms-priority-intermediate	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p36-5-day-professional-compound-intermediate	3
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p28-5-day-split-weak-point-intermediate	4
RECOVERY_SPACING_INVALID REQUIRED_MOVEMENT_PATTERN_MISSING	rejected	0	0	0	0	0	0	0	p30-5-day-upper-priority-iranian-intermediate	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p32-5-day-upper-lower-specialty-intermediate	6

p32-5-day-upper-lower-specialty-intermediate → competition:5:intermediate:60

Professional candidates before first Upper/Lower rank: 4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p28-5-day-split-weak-point-intermediate	1
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p26-5-day-classic-body-part-intermediate	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p34-5-day-fst7-arms-priority-intermediate	3
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p36-5-day-professional-compound-intermediate	4
RECOVERY_SPACING_INVALID REQUIRED_MOVEMENT_PATTERN_MISSING	rejected	0	0	0	0	0	0	0	p30-5-day-upper-priority-iranian-intermediate	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p32-5-day-upper-lower-specialty-intermediate	6

p32-5-day-upper-lower-specialty-intermediate → competition:5:intermediate:75

Professional candidates before first Upper/Lower rank: 4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p28-5-day-split-weak-point-intermediate	1
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p26-5-day-classic-body-part-intermediate	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p34-5-day-fst7-arms-priority-intermediate	3
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p36-5-day-professional-compound-intermediate	4
RECOVERY_SPACING_INVALID REQUIRED_MOVEMENT_PATTERN_MISSING	rejected	0	0	0	0	0	0	0	p30-5-day-upper-priority-iranian-intermediate	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p32-5-day-upper-lower-specialty-intermediate	6

p32-5-day-upper-lower-specialty-intermediate → competition:5:intermediate:90

Professional candidates before first Upper/Lower rank: 4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p28-5-day-split-weak-point-intermediate	1
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p26-5-day-classic-body-part-intermediate	2
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p34-5-day-fst7-arms-priority-intermediate	3
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p36-5-day-professional-compound-intermediate	4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID REQUIRED_MOVEMENT_PATTERN_MISSED SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	0	0	0	0	0	0	0	p30-5-day-upper-priority-iranian-intermediate	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p32-5-day-upper-lower-specialty-intermediate	6

p33-5-day-upper-lower-specialty-advanced → competition:5:advanced:30

Professional candidates before first Upper/Lower rank: 4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p29-5-day-split-weak-point-advanced	1
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p37-5-day-professional-compound-advanced	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p27-5-day-classic-body-part-advanced	3
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p35-5-day-fst7-arms-priority-advanced	4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	0	0	0	0	0	0	0	p31-5-day-upper-priority-iranian-advanced	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p33-5-day-upper-lower-specialty-advanced	6

p33-5-day-upper-lower-specialty-advanced → competition:5:advanced:45

Professional candidates before first Upper/Lower rank: 4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p27-5-day-classic-body-part-advanced	1
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p35-5-day-fst7-arms-priority-advanced	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p29-5-day-split-weak-point-advanced	3
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p37-5-day-professional-compound-advanced	4
RECOVERY_SPACING_INVALID	rejected	0	0	0	0	0	0	0	p31-5-day-upper-priority-iranian-advanced	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p33-5-day-upper-lower-specialty-advanced	6

p33-5-day-upper-lower-specialty-advanced → competition:5:advanced:60

Professional candidates before first Upper/Lower rank: 4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p29-5-day-split-weak-point-advanced	1
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p37-5-day-professional-compound-advanced	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p27-5-day-classic-body-part-advanced	3
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p35-5-day-fst7-arms-priority-advanced	4
RECOVERY_SPACING_INVALID	rejected	0	0	0	0	0	0	0	p31-5-day-upper-priority-iranian-advanced	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p33-5-day-upper-lower-specialty-advanced	6

p33-5-day-upper-lower-specialty-advanced → competition:5:advanced:75

Professional candidates before first Upper/Lower rank: 4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p29-5-day-split-weak-point-advanced	1
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p37-5-day-professional-compound-advanced	2
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p27-5-day-classic-body-part-advanced	3
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p35-5-day-fst7-arms-priority-advanced	4
RECOVERY_SPACING_INVALID	rejected	0	0	0	0	0	0	0	p31-5-day-upper-priority-iranian-advanced	5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p33-5-day-upper-lower-specialty-advanced	6

p33-5-day-upper-lower-specialty-advanced → competition:5:advanced:90

Professional candidates before first Upper/Lower rank: 4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p29-5-day-split-weak-point-advanced	1
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p37-5-day-professional-compound-advanced	2
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p27-5-day-classic-body-part-advanced	3

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p35-5-day-fst7-arms-priority-advanced	4
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	0	0	0	0	0	0	0	p31-5-day-upper-priority-iranian-advanced	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p33-5-day-upper-lower-specialty-advanced	6

p40-6-day-upper-lower-x3-intermediate → competition:6:intermediate:30

Professional candidates before first Upper/Lower rank: 5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p46-6-day-classic-body-part-intermediate	1
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p44-6-day-arnold-split-intermediate	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p48-6-day-ronnie-double-exposure-intermediate	3

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p42-6-day-fitclub-hybrid-intermediate	4
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p38-6-day-ppl-ab-intermediate	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p40-6-day-upper-lower-x3-intermediate	6

p40-6-day-upper-lower-x3-intermediate → competition:6:intermediate:45

Professional candidates before first Upper/Lower rank: 5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p46-6-day-classic-body-part-intermediate	1
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p48-6-day-ronnie-double-exposure-intermediate	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p44-6-day-arnold-split-intermediate	3
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p38-6-day-ppl-ab-intermediate	4
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p42-6-day-fitclub-hybrid-intermediate	5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p40-6-day-upper-lower-x3-intermediate	6

p40-6-day-upper-lower-x3-intermediate → competition:6:intermediate:60

Professional candidates before first Upper/Lower rank: 5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_EXERCISE_COUNT_OUT_OF_RANGE SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p46-6-day-classic-body-part-intermediate	1
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p44-6-day-arnold-split-intermediate	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p48-6-day-ronnie-double-exposure-intermediate	3
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p42-6-day-fitclub-hybrid-intermediate	4
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p38-6-day-ppl-ab-intermediate	5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p40-6-day-upper-lower-x3-intermediate	6

p40-6-day-upper-lower-x3-intermediate → competition:6:intermediate:75

Professional candidates before first Upper/Lower rank: 5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED SESSION_EXERCISE_COUNT_OUT_OF_RANGE RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p46-6-day-classic-body-part-intermediate	1
SEMANTIC_OPENER_CONFLICT RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p44-6-day-arnold-split-intermediate	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p48-6-day-ronnie-double-exposure-intermediate	3
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	40	40	0	0	0	0	0	p42-6-day-fitclub-hybrid-intermediate	4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p38-6-day-ppl-ab-intermediate	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p40-6-day-upper-lower-x3-intermediate	6

p41-6-day-upper-lower-x3-advanced → competition:6:advanced:45

Professional candidates before first Upper/Lower rank: 5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p47-6-day-classic-body-part-advanced	1
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p45-6-day-arnold-split-advanced	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p49-6-day-ronnie-double-exposure-advanced	3
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p39-6-day-ppl-ab-advanced	4
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p43-6-day-fitclub-hybrid-advanced	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p41-6-day-upper-lower-x3-advanced	6

p41-6-day-upper-lower-x3-advanced → competition:6:advanced:60

Professional candidates before first Upper/Lower rank: 5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p47-6-day-classic-body-part-advanced	1
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p49-6-day-ronnie-double-exposure-advanced	2
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p45-6-day-arnold-split-advanced	3
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p43-6-day-fitclub-hybrid-advanced	4
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p39-6-day-ppl-ab-advanced	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p41-6-day-upper-lower-x3-advanced	6

p41-6-day-upper-lower-x3-advanced → competition:6:advanced:75

Professional candidates before first Upper/Lower rank: 5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p47-6-day-classic-body-part-advanced	1
RECOVERY_SPACING_INVALID	rejected	50	50	0	0	0	0	0	p49-6-day-ronnie-double-exposure-advanced	2
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p45-6-day-arnold-split-advanced	3
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	40	40	0	0	0	0	0	p43-6-day-fitclub-hybrid-advanced	4
RECOVERY_SPACING_INVALID	rejected	40	40	0	0	0	0	0	p39-6-day-ppl-ab-advanced	5
TEMPLATE_ATTEMPT_SUCCEEDED	succeeded	0	0	0	0	0	0	0	p41-6-day-upper-lower-x3-advanced	6

dynamic:upper_lower_x3 → competition:6:advanced:90

Professional candidates before first Upper/Lower rank: 5

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p47-6-day-classic-body-part-advanced	1
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p49-6-day-ronnie-double-exposure-advanced	2
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	50	50	0	0	0	0	0	p45-6-day-arnold-split-advanced	3
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	40	40	0	0	0	0	0	p43-6-day-fitclub-hybrid-advanced	4

Reasons	Result	total	professional	fallback	sex	goal	body	priority	Template	Rank
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED RECOVERY_SPACING_INVALID SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	40	40	0	0	0	0	0	p39-6-day-ppl-ab-advanced	5
SESSION_DURATION_UNDER_TARGET SESSION_DURATION_TARGET_UNSATISFIED SESSION_DURATION_CONSTRAINT_UNEXPLAINED	rejected	0	0	0	0	0	0	0	p41-6-day-upper-lower-x3-advanced	6

Duration Deep Dive .6

Duration failures	Success	FAIL	PASS	Tests	Duration
0	36.1%	23	13	36	min 30
0	41.7%	21	15	36	min 45
6	41.7%	21	15	36	min 60
11	44.4%	20	16	36	min 75
26	27.8%	26	10	36	min 90

Count	Repair diagnostic
47	Duration/count failure cases
1046/1673	Set Addition success/calls
503/751	Exercise Addition success/calls
wrong_template_muscle:91341 weekly_hard_volume_cap:31852 no_useful_candidate:25504 acceptable_volume_preference:15391 duplicate:5115 exercise_set_cap:3645 near_equivalent:2245 session_volume_cap:363	Rejected candidate categories

در این snapshot، Duration failure در 30 و 45 دقیقه علت اصلی نبود؛ از 60 دقیقه شروع شد و در 90 دقیقه به 26 مورد رسید. Repair فقط می‌تواند work امن و مفید را تا سقف‌های weekly hard count، session volume، target muscle و volume، semantic equivalence اضافه کند.

Topology Comparison .7

Validation/Gate	Construction	Recovery	Volume	Duration	Success	FAIL	PASS	Tests	Topology
15	4	10	0	7	68.3%	19	41	60	Upper/Lower
10	0	10	0	4	50.0%	10	10	20	PPL
52	0	49	0	22	20.0%	52	13	65	Body-Part
11	0	10	0	4	26.7%	11	4	15	Arnold-style
10	0	10	0	4	0.0%	10	0	10	Specialization
9	0	9	0	2	10.0%	9	1	10	PPL×2

Root Cause Ranking .8

Function	Relevant file	Cases	Root cause	Rank
recovery_spacing_is_valid() / repair_recovery_weekdays()	app/workouts/program_engine/ recovery.py	62	RECOVERY_SPACING_INVALID	1
repair_session_durations() / _repair_underfill()	app/workouts/program_engine/ session_duration.py	42	SESSION_DURATION_UNDER_TARGET	2
validate_program()	app/workouts/program_engine/ validation.py	4	TEMPLATE_SESSION_EXERCISE_COUNT_UN SATISFIED	3
validate_program()	app/workouts/program_engine/ validation.py	2	REQUIRED_MOVEMENT_PATTERN_MISSING	4
repair_session_durations() / _repair_underfill()	app/workouts/program_engine/ session_duration.py	1	SESSION_DURATION_TARGET_UNSATISFIE D	5

9. Architectural Diagnosis — بدون پیشنهاد Fix

(1 Professional preference یک ordering preference است، نه survival contract

رفتار منطقی؟ بله؛ preference نباید hard safety/quality gates را دور بزند. **False negative**؟ خود score نه، اما feasibility آن فقط slot/capacity اولیه را می‌سنجد و نتیجه‌ی post-prescription recovery/volume/validation را پیش‌بینی نمی‌کند. **Topology impact**: Template حرفه‌ای rank بالا می‌گیرد، fail می‌شود و Upper/Lower کم‌امتیاز بعدی زنده می‌ماند. شاهد: Competition output 17 از 30.

(2 Recovery spacing بزرگ‌ترین survival filter است

رفتار منطقی؟ حفاظت از دو exposure متوسط/سنگین مجاور منطقی است. **False negative risk**: بالا؛ validator تمام secondary set ها را با ضریب ثابت 0.5 در exposure load حساب می‌کند و برای 5/6 روز، تعداد weekday arrangement ممکن محدود است. **Topology impact**: Body-Part و PPL/Arnold روزهای هم‌پوشان push/pull/arms را جدا می‌کنند؛ Upper/Lower هم‌پوشانی را داخل همان session جمع می‌کند و از pairهای مجاور کمتری عبور می‌کند. شاهد: Body-Part recovery failures=49 و Upper/Lower recovery failures=10.

(3 Duration underfill با افزایش duration تشدید می‌شود

رفتار منطقی؟ قرارداد ± 10 requested و count حداقل 5 برای +45 سخت و صریح است. **False negative risk**: متوسط تا بالا برای body-part/specialization؛ یک روز عضله‌ای باریک ممکن است Candidate مفید و غیرتکراری کافی برای 80 دقیقه main training نداشته باشد. **Structural advantage**: Upper/Lower target-muscle pool بزرگ‌تری دارد و repair گزینه‌های بیشتری می‌بیند. شاهد: 90min duration failures=26.

(4 minute count/construction failures-30 محدود اما قطعی‌اند

رفتار منطقی؟ بله، سقف 3-4 MAIN برای 30 دقیقه hard invariant است. **False negative risk**: وقتی core template slots پس از trim هنوز از سقف عبور کنند، این failure قبل از volume/duration repair رخ می‌دهد و در چهار case دیده شد.

(5 برخی topologyها عملاً survival بسیار پایین دارند

Specialization survival=0.0% و PPLx2 survival=10.0%. این نتیجه از Baseline بدون priority است؛ بنابراین specialization bonus تطبیقی فعال نبود. این Audit ثابت نمی‌کند Template ها برای profile اولویت‌دار نیز همین نرخ را دارند، اما mechanics پایه را نشان می‌دهد.

Diagnosis نهایی: Upper/Lower به‌خاطر score برنده نمی‌شود؛ به‌خاطر survival advantage پس از fail شدن topologyهای حرفه‌ای برنده می‌شود. در این snapshot، عامل غالب recovery spacing و سپس duration underfill است. این گزارش هیچ Implementation change پیشنهاد یا اعمال نمی‌کند.

محدودیت Audit

این ماتریس mechanics را با دو Baseline کنترل‌شده می‌سنجد، نه distribution واقعی کاربران. Equipment و injury عملاً حذف شدند. Candidate rejection category های duration از wrapper تشخیصی همان run گرفته شده‌اند؛ آن‌ها تعداد ارزیابی/رد را نشان می‌دهند و می‌توانند برای یک Candidate بیش از یک constraint ثبت کنند.